

The coppe document class

Version 3.8

Vicente H. F. Batista George O. Ainsworth Jr.
Geraldo Xexéo Eduardo Mangeli

May 27, 2026

Abstract

In this work, it is described the `coppe` document class as well as other files distributed by the COPPETEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

1 Introduction

Writing documents in L^AT_EX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typesetted in L^AT_EX. This demand motivated the creation of the COPPETEX project, which tries to facilitate and encourage the use of L^AT_EX within the COPPE/UFRJ scope.

The `coppe` document class is the main product of COPPETEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the `coppe` class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The COPPETEX project comprehends also BibTEX and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the `coppe` class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

2 License

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copying, distributing or modifying the source code, among other acts covered by this license.

To see the full text of the GNU GPL license, go to the `COPYING` file attached to this package.

3 Support

We maintain a mailing list where users can send questions, comments, and bugs to. More details can be found [here](#).

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to <https://github.com/COPPE-UFRJ/CoppeTeX>.

4 User interface

`\frontmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain
`\mainmatter` three-parts: *front*, *main* and *back* matters [?]. Each one of these parts is started
`\backmatter` by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`.
The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

Cover (capa)

The ABNT cover suggested in the manual's Annex A. The `\maketitle` command builds it automatically, ahead of the title page: the university, the full COPPE name, and the graduate program (taken from `\department`), followed by the author, the title, and the city and year, all centred and in uppercase. The UFRJ and COPPE logos head the page. The capa is not counted.

Front face

The front face is unnumbered. There, it is not allowed to use hyphenation [?]. It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` The macros `\title` and `\foreigntitle` are used to enter the titles of your
`\foreigntitle` monograph in the current and foreign languages. The default languages are Brazilian Portuguese and English. The `babel` package is automatically loaded by `coppe.cls`, so you do not need to load it again. The Brazilian Portuguese is the main language and the English is only required for the foreign abstract.

`\advisor` Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc.
`\examiner`

students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command `\advisor` as below:

```
\advisor{Title}{Advisor's Name}{Surname}{Degree}
\advisor{Title}{Second Advisor's Name}{Surname}{Degree}
\advisor{Title}{Third Advisor's Name}{Surname}{Degree}
```

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the `\examiner` macro. The examiners' names are entered differently:

```
\examiner{Title}{First Examiner's Name Surname}{Degree}
\examiner{Title}{Second Examiner's Name Surname}{Degree}
...
\examiner{Title}{N-th Examiner's Name Surname}{Degree}
```

Remember that all names must be given before calling `\maketitle`.

`\department` The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENO), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., `\department{PEC}`.

`\date` This macro is used to set the month and year of defense. This information is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as `\date{10}{2007}`.

`\keyword` The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call `\maketitle`. It is not needed in qualifying exams, though.

Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

Abstracts

`abstract (env.)` As stated by the Academic Registry [?], abstracts must be in one page each, with
`foreignabstract (env.)` at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

Lists of symbols and abbreviations (optional)

`\abbrev` The lists of symbols and abbreviations are optional, although highly recommended.
`\syml` It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\syml[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.
`\makeloabbreviations` These lists are lexicographically sorted by using the MakeIndex program, which
`\makelosymbols` is part of any L^AT_EX implementation. For `\syml`, if the optional parameter is
`\printloabbreviations` provided, it will be used as sort key. This was later, in 2024, implemented also for
`\printlosymbols` `\abbrev`, otherwise `Symbol`, or `Abbreviation` will be used as sort key, what can result in an undesirable order if it contains L^AT_EX commands, mathematical symbols, or mix of uppercase and lowercase. MakeIndex needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the `coppe` class provides the commands `\makeloabbreviations` and `\makelosymbols`, respectively. They must be called in the document preamble. The commands `\printlosymbols` and `\printloabbreviations` have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call `latex`, it will be created two files with extensions `abx` and `syx`, which contain MakeIndex input data. They must be processed with `makeindex` in order to get the lists correctly produced, redirecting the output to files with extension `lab` and `los` respectively:

```
makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx
```

Note the `-s` option for specifying the style `coppe.ist`. Now, rerun `latex` twice to get the references solved and you are done.

References

COPPE_T_EX produces citations and the reference list with `biblatex` and the `biber` backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no `\bibliographystyle` to set and no Bib_T_EX `.bst` file to choose (the former `coppe-plain` and `coppe-unsrt` Bib_T_EX styles have been retired).

By default, citations follow the author–date style, e.g., “Chartier (2002)”. Passing the `numbers` class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with `\addbibresource` and print the list where it should appear with `\printbibliography`:

```
\addbibresource{example.bib} % in the preamble
```

```
...
\printbibliography           % where the list goes
```

Entry types and fields may be written in English (`@book`, `title`, `author`) or with their unaccented Portuguese synonyms (`@livro`, `titulo`, `autor`).

Run in sequence `LATEX`, `biber`, and twice again `LATEX` to resolve the citations and references. The style is `natbib` compatible (via the `biblatex natbib=true` option), so you may freely issue `\citet` and `\citep`, as well as the other `natbib` commands.

Appendix and Annex (Optional)

`\appendix` Appendices and annexes are optional chapters that are part of the back matter.

`\annex` The `\appendix` command is a standard `LATEX` command used before all appendices. `COPPETEX` introduces the `\annex` command, which should be used only in the back matter (i.e., after the `\backmatter` command) and only after all existing `\appendix` chapters. This restriction is due to implementation constraints.

Therefore, the order for backmatter is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the `coppe` class. These options can be passed to `coppe` as follows: `\documentclass[option1, option2]{coppe}`. In which follows, we give a brief description of all supported options.

dsc, **msc**, **dscexam**, **mscexam** The `coppe` class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the `dsc`, `msc`, `mscexam`, and `dscexam` options, respectively.

doublespacing The default line spacing is one-and-a-half. For enabling double spacing between lines, use the `doublespacing` option.

numbers The default citation style is the author–date scheme. For numbered citations, specify the `numbers` option to the `coppe` class; the reference list is then numbered in order of citation (unsorted). The matching `biblatex` style is selected automatically in either case.

`english` COPPET_{EX} uses Babel. The default language is Portuguese (actually `brazilian`), with English being the second language. If option `english` is used, English becomes the main language and Portuguese the secondary. Look at the Babel package to switch between languages.

5.1 Changing document identification

`\freeconfig` The user could *optionally* use the command `freeconfig` to modify the parameters that print the document identification. The command `freeconfig` needs all those parameters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

6 Quick, useful tips

Pictures. The default picture format of L_AT_EX is the Encapsulated PostScript (EPS). If you use pdfL_AT_EX, the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and pdf_lat_ex will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available [here](#).

Fonts. The default font in L_AT_EX is the Computer Modern. If you would like to try its enhanced version, consider using the `lmodern` package. To use Times, it is recommended to load the package `mathptmx`, rather than the deprecated `times`. There is also an enhanced Times version available with the `tgtermes` package. You can still use the Arial font face with the package `uarial`.

Hyperref. When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the `hyperref` package. It is also useful to enable hyperlinks. Fortunately, the `coppe` class will do this automatically if `hyperref` is loaded.

Printing. To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

`longquote` **Quotation** To quote text larger than three lines, according to ABNT, you must increase the left margin to 4 cm, do not use quotation marks, and use a smaller font. The `coppe` class provides the `longquote` environment to easily make these adjustments.

7 A simple example

```
1 \example
2 \documentclass[dsc]{coppe}
3
4 \usepackage{booktabs}% tabelas mais bonitas
5 \usepackage{rotating}% rodando coisas, como tabelas
6 \usepackage{longtable} % tabelas longas
7 \usepackage[most]{tcolorbox} % caixas de texto
8 \usepackage{amsmath,amssymb}
9 \usepackage{hyperref}
10 % listings, fancyvrb e algorithm2e já vêm carregados pela classe coppe (v3.8).
11
12 \addbibresource{example.bib}% base de referências (biblatex/biber)
13
14 \makelosymbols
15 \makeloabbreviations
16
17 \begin{document}
18   \title{Titulo da Tese}
19   \foreigntitle{Thesis Title}
20   \author{Nome do Autor}{Sobrenome}
21   \advisor{Prof.}{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}
22   \advisor{Prof.}{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}
23   \advisor{Prof.}{Nome do Terceiro Orientador}{Sobrenome}{D.Sc.}
24
25   \examiner{Prof.}{Nome do Primeiro Examinador Sobrenome}{D.Sc.}
26   \examiner{Prof.}{Nome do Segundo Examinador Sobrenome}{Ph.D.}
27   \examiner{Prof.}{Nome do Terceiro Examinador Sobrenome}{D.Sc.}
28   \examiner{Prof.}{Nome do Quarto Examinador Sobrenome}{Ph.D.}
29   \examiner{Prof.}{Nome do Quinto Examinador Sobrenome}{Ph.D.}
30   \department{PESC}
31   \date{01}{2024}
32
33   \keyword{Primeira palavra-chave}
34   \keyword{Segunda palavra-chave}
35   \keyword{Terceira palavra-chave}
36
37   \maketitle
38
39   \frontmatter
40   \dedication{A alguém cujo valor é digno desta dedicatória.}
41
42   \chapter*{Agradecimentos}
43
44   Gostaria de agradecer a todos os que contribu\iram para este trabalho:
45   orientadores, colegas, familiares e \as ag\encias de fomento. Lorem ipsum
46   dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
47   ut labore et dolore magna aliqua.
48
49   \begin{abstract}
50
51   Apresenta-se, nesta tese, um exemplo de resumo. Lorem ipsum dolor sit amet,
52   consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
```

```

53 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
54 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
55 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
56 pariat. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
57 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
58 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
59 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
60 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
61 aspernatur aut odit aut fugit.
62
63 \end{abstract}
64
65 \begin{foreignabstract}
66
67 In this work, we present an example abstract. Lorem ipsum dolor sit amet,
68 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
69 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
70 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
71 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
72 pariat. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
73 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
74 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
75 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
76 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
77 aspernatur aut odit aut fugit.
78
79 \end{foreignabstract}
80
81 % Ordem pre-textual (NBR 14724/6027, R55): listas (ilustracoes, tabelas,
82 % abreviaturas e siglas, simbolos) ANTES do sumario; sumario por ultimo.
83 \listoffigures
84 \listoftables
85 \listofquadros
86 \listofprogramas
87 \listofalgorithms
88 \printloabbreviations
89 \printlosymbols
90 \tableofcontents
91
92 \mainmatter
93 \chapter{Introdução}
94
95 Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos da
96
97 A classe \verb|coppe| foi criada por Vicente Helano e George Ainsworth, porém, em 2024, é
98
99 A versão mais atual dessa classe é mantida no GitHub, no repositório \url{https://github.com}
100
101 Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po
102
103 Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p
104
105 É importante de notar que essa classe não foi construída sobre a classe \LaTeX \ para a A
106

```


107 Apesar desse modelo ser muito bom, ele tem um defeito: a limitação do sistema de referências
108
109 Mais ainda, as regras da COPPE ainda não se adaptaram, no início de 2024, as novas regras
110
111 Este documento não substitui, mas complementa, o documento que descreve a classe.
112
113
114
115 \chapter{Configurações Iniciais}
116
117 A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no comando
118
119 Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o traçado
120
121 Recomendo ler o documento “‘The \verb*|coppe| document class’” para entender melhor todas as opções
122
123 \section{Linguagem principal do texto}
124
125 Essa classe considera que o texto principal está em português e algumas partes específicas
126 \begin{itemize}
127 \item A opção \verb|english| deve ser usada no comando \verb|\documentclass|.
128 \item A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é possível
129 \end{itemize}
130
131 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe \CopeTeX
132
133
134 \section{Por que usar o \LaTeX}
135
136 Há uma grande discussão entre usuário de Word e \LaTeX, principalmente, quanto ao uso desses programas
137
138 Nós escolhemos o \LaTeX por alguns motivos: grande facilidade de seguir um estilo sem se preocupar
139
140 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sobre o documento
141
142 \section{Como e onde usar o \LaTeX}
143
144 Existem muitos tutoriais de \LaTeX, mas basicamente, em 2024, ele é usado em dois ambientes
145 \begin{enumerate}
146 \item Na sua máquina, instalando uma versão completa como o \hologo{MiKTeX}, típico do Windows
147 \item Usar um ambiente na rede, como o Overleaf.
148 \end{enumerate}
149
150 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o backup
151
152
153 \chapter{Algumas Regras da COPPE}
154
155 Todas abreviaturas e símbolos devem ser definidas antes de utilizadas. Isso é facilmente feito
156
157 É imprescindível definir os símbolos, tal como o
158 conjunto dos números reais $\mathbb{R}$ e o conjunto vazio $\emptyset$.
159 \syml{\mathbb{R}}{Conjunto dos números reais}
160 \syml{\emptyset}{Conjunto vazio}. Usamos esse exemplo aqui justamente para mostrar como

161

162 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o `\ve`

163

164 Como as listas de símbolos e de abreviaturas usamo o mesmo comando usado para criar índices

165

166 `\section{Citações}`

167

168 Citações curtas podem ser feitas `\quote{o comando quote}` ou direto com “duas crases e dois

169

170 `\begin{longquote}`

171 Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)

172 feita com o ambiente `\verb=longquote=` The primary objective of this

173 investigation was to determine the feasibility of detecting corrosion in

174 aluminum Naval aircraft components with neutron radiographic interrogation

175 and the use of standard corrosion penetrameters. Secondary objectives

176 included the determination of the effect of object thickness on image quality,

177 the defining of minimum levels of detectability and a preliminary investigation

178 of a means whereby the degree of corrosion could be quantified with neutron

179 radiographic data. `\cite{article-example}`

180 `\end{longquote}`

181

182 Citações devem apontar as referências. Para isso, está disponível o ótimo pacote `\verb*|na`

183

184 Em todo caso, `\textbf{deve}` se tomar enorme atenção com as citações, para evitar ocorrer em

185

186 Quando se cita uma obra por meio de outra (`\textit{apud}`), pela ABNT entra na lista de ref

187

188 `\chapter{Floats}`

189

190 Grande parte dos problemas de iniciantes, e veteranos, em `\LaTeX` é da localização dos `\texti`

191

192 A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesm

193

194 `\textbf{Pela norma atual (manual2024), a legenda}` `\verb|\caption|` `\textbf{de toda ilustração}`

195

196 Quadros são ilustrações cujos dados são delimitados por linhas em `\emph{todas}` as bordas (ao

197

198

199 `\section{Tabelas e Figuras Padrão}`

200

201 Vamos ver uma tabela padrão, como a `\autoref{tab:exemplo_numeros}`.

202

203 `\begin{table}[ht]`

204 `\centering` % Centraliza a tabela

205 `\caption{Exemplo de Tabela de Números}`

206 `\label{tab:exemplo_numeros}`

207 `\begin{tabular}{ccc}` % Define a quantidade de colunas

208 `\hline` % Linha superior

209 `\textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\` % Cabeçalhos

210 `\hline` % Linha média

211 `1 & 2 & 3 \\` % Primeira linha de dados

212 `\hline`

213 `4 & 5 & 6 \\` % Segunda linha de dados

214 `\hline`

```

215 7 & 8 & 9 \\ % Terceira linha de dados
216 \hline
217 10 & 11 & 12 \\ % Quarta linha de dados
218 \hline % Linha inferior
219 \end{tabular}
220 \fonte{Elabora\c{c}\~ao pr\'opria.} % fonte obrigatoria, abaixo do float
221 \end{table}
222
223
224
225 Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.
226
227 \begin{figure}[ht]
228 \centering % Centraliza a figura
229 \caption{Exemplo de Figura com Legenda Acima} % Legenda em cima (norma atual)
230 \label{fig:exemplo_figura} % Etiqueta para referência cruzada
231 \includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largura
232 \source{COPPE/UFRJ.} % fonte obrigatoria, abaixo da figura
233 \end{figure}
234
235 Quadros têm a mesma regra (legenda acima, fonte abaixo), mas com linhas em todas as bordas.
236
237 \begin{quadro}[ht]
238 \centering
239 \caption{Exemplo de Quadro}
240 \label{qua:exemplo}
241 \begin{tabular}{|l|l|}
242 \hline
243 \textbf{Ilustração} & \textbf{Delimitação} \\
244 \hline
245 Tabela & Linhas apenas no topo e na base. \\
246 \hline
247 Quadro & Linhas em todas as bordas. \\
248 \hline
249 \end{tabular}
250 \fonte{Elabora\c{c}\~ao pr\'opria.}
251 \end{quadro}
252
253
254
255
256
257 \section{Tabelas mais elegantes}
258
259 Atualmente a tendência é usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Iss
260
261 \begin{table}[ht]
262 \centering % Centraliza a tabela
263 \caption{Exemplo de Tabela de Números mais elegantes}
264 \label{tab:exemplo_numerosbom}
265 \begin{tabular}{ccc} % Define a quantidade de colunas
266 \toprule % Linha superior
267 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\
268 \midrule % Linha média

```

```

269 1 & 2 & 3 \\ % Primeira linha de dados
270 4 & 5 & 6 \\ % Segunda linha de dados
271 7 & 8 & 9 \\ % Terceira linha de dados
272 10 & 11 & 12 \\ % Quarta linha de dados
273 \bottomrule % Linha inferior
274 \end{tabular}
275 \end{table}
276
277 \section{Tabelas Longas ou Largas}
278
279 Se sua tabela é muito longa ou larga, existem várias opções.
280 \begin{itemize}
281     \item alterar o tamanho da letra
282     \item Usar o longtable
283     \item rodar a tabela, fazendo ela em \textit{landscape}
284     \item fazer a tabela dentro de um minibox
285 \end{itemize}
286
287
288 \subsection{Tabelas largas demais}
289
290 É comum em teses que as tabelas sejam largas demais. Há várias formas de resolver isso.
291
292 A \autoref{tab:tabela_largafns} é larga demais, e nela isso é resolvido diminuindo a fonte p
293
294 \begin{table}[ht]
295 \centering % Centraliza a tabela
296 \caption{Exemplo de Tabela Larga com Fonte Menor}
297 \label{tab:tabela_largafns}
298 \footnotesize % Aplica uma fonte menor para a tabela
299 \begin{tabular}{ccccccc} % Aumente o número de colunas conforme necessário
300 \toprule
301 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} & \textbf{Coluna 4} & \textbf{Coluna 5} & \textbf{Coluna 6} & \textbf{Coluna 7} \\
302 \midrule
303 Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\
304 Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
305 Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
306 \bottomrule
307 \end{tabular}
308 \end{table}
309
310 O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer co
311
312 \begin{table}[ht]
313 \centering
314 \caption{Exemplo de Tabela Redimensionada}
315 \label{tab:examplerb}
316 \resizebox{\textwidth}{!}{%
317 \begin{tabular}{llll}
318 \toprule
319 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 \\
320 \midrule
321 Dados 1 & Dados 2 & Dados 3 & Dados 4 \\
322 Dados 5 & Dados 6 & Dados 7 & Dados 8 \\

```

```

323 \bottomrule
324 \end{tabular}%
325 }
326 \end{table}
327
328
329 Para rodar uma tabela muito larga em 90 graus no LaTeX, você pode usar o pacote \verb*|rotat
330
331 Aqui está um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela. Pr
332
333 \begin{sidewaystable}
334 \centering
335 \caption{Sua Legenda Aqui}
336 \label{tab:sua_tabela}
337 \begin{tabular}{lll}
338 \toprule
339 Coluna 1 & Coluna 2 & Coluna 3 \\
340 \midrule
341 Item 1 & Item 2 & Item 3 \\
342 Item 4 & Item 5 & Item 6 \\
343 \bottomrule
344 \end{tabular}
345 \end{sidewaystable}
346
347 Se a tabela for muito longa, o ambiente \verb|longtable| é o ideal. Ele fornece comandos par
348
349 % Exemplo de tabela longa que se estende por várias páginas
350 \begin{longtable}{|c|c|c|}
351 % primeiro cabeçalho (é o caption)
352 \caption{Exemplo de Tabela Longa}\label{tab:longa} \\
353 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline
354 \endfirsthead
355 % cabeçalho normal
356 \multicolumn{3}{c}%
357 {\table\ -- continuação da página anterior} \\
358 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline
359 \endhead
360 % pé normal
361 \hline \multicolumn{3}{r|}{Continua na próxima página} \\ \hline
362 \endfoot
363 \hline
364 % último pé
365 \multicolumn{3}{r|}{Continua na próxima página} \\
366 \hline \hline
367 \endlastfoot
368
369 % Conteúdo da tabela
370 1 & 2 & 3 \\
371 4 & 5 & 6 \\
372 1 & 2 & 3 \\
373 4 & 5 & 6 \\
374 1 & 2 & 3 \\
375 4 & 5 & 6 \\
376 1 & 2 & 3

```

377 4 & 5 & 6 \\
 378 1 & 2 & 3 \\
 379 4 & 5 & 6 \\
 380 1 & 2 & 3 \\
 381 4 & 5 & 6 \\
 382 1 & 2 & 3 \\
 383 4 & 5 & 6 \\
 384 1 & 2 & 3 \\
 385 4 & 5 & 6 \\
 386 1 & 2 & 3 \\
 387 1 & 2 & 3 \\
 388 4 & 5 & 6 \\
 389 1 & 2 & 3 \\
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 403 1 & 2 & 3 \\
 404 4 & 5 & 6 \\
 405 1 & 2 & 3 \\
 406 4 & 5 & 6 \\
 407 1 & 2 & 3 \\
 408 4 & 5 & 6 \\
 409 1 & 2 & 3 \\
 410 4 & 5 & 6 \\\ 1 & 2 & 3 \\
 411 4 & 5 & 6 \\
 412 1 & 2 & 3 \\
 413 4 & 5 & 6 \\
 414 1 & 2 & 3 \\
 415 1 & 2 & 3 \\
 416 4 & 5 & 6 \\
 417 1 & 2 & 3 \\
 418 4 & 5 & 6 \\
 419 1 & 2 & 3 \\
 420 4 & 5 & 6 \\
 421 1 & 2 & 3 \\
 422 4 & 5 & 6 \\
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 424 4 & 5 & 6 \\
 425 1 & 2 & 3 \\
 426 4 & 5 & 6 \\
 427 1 & 2 & 3 \\
 428 4 & 5 & 6 \\
 429 1 & 2 & 3 \\
 430 4 & 5 & 6 \\

```

431 1 & 2 & 3 \\
432 4 & 5 & 6 \\
433 1 & 2 & 3 \\
434 4 & 5 & 6 \\
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466 4 & 5 & 6 \\
467 1 & 2 & 3 \\
468 4 & 5 & 6 \\
469 1 & 2 & 3 \\
470 4 & 5 & 6 \\
471 1 & 2 & 3 \\
472 4 & 5 & 6 \\
473 1 & 2 & 3 \\
474 4 & 5 & 6 \\
475 1 & 2 & 3 \\
476 4 & 5 & 6 \\
477
478 % Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas
479 \end{longtable}
480
481 \chapter{Revis\~ao Bibliogr\'afica}
482
483 Para ilustrar a completa ades\~ao ao estilo de cita{\c c}\~oes e listagem de
484 refer\~encias bibliogr\'aficas, a Tabela~\ref{tab:citation} apresenta cita{\c

```

```

485 c}\~oes de alguns dos trabalhos contidos na norma fornecida pela CPGP da
486 COPPE, utilizando o estilo numérico. Tirando do comando inicial o parâmetro opcional numér
487
488 \begin{table}[h]
489 \caption{Exemplos de cita{\c c}\~oes utilizando o comando padr\~ao
490 \texttt{\textbackslash cite} do \LaTeX\ e
491 o comando \texttt{\textbackslash citet},
492 fornecido pelo pacote \texttt{natbib}.}
493 \label{tab:citation}
494 \centering
495 {\footnotesize
496 \begin{tabular}{|c|c|c|}
497 \hline
498 Tipo da Publicação & \verb|\cite| & \verb|\citet|\\
499 \hline
500 Livro & \cite{book-example} & \citet{book-example}\\
501 Artigo & \cite{article-example} & \citet{article-example}\\
502 Relatório & \cite{techreport-example} & \citet{techreport-example}\\
503 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\\
504 Anais de Congresso & \cite{inproceedings-example} &
505 \citet{inproceedings-example}\\
506 Séries & \cite{incollection-example} & \citet{incollection-example}\\
507 Em Livro & \cite{inbook-example} & \citet{inbook-example}\\
508 Dissertação de mestrado & \cite{mastersthesis-example} &
509 \citet{mastersthesis-example}\\
510 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\\
511 \hline
512 \end{tabular}}
513 \end{table}
514
515 \begin{table}[h]
516 \caption{Exemplos de cita{\c c}\~oes utilizando o comando padr\~ao
517 \texttt{\textbackslash cite} do \LaTeX\ e
518 o comando \texttt{\textbackslash citet},
519 fornecido pelo pacote \texttt{natbib}. Além disso, usando o booktabs.}
520 \label{tab:citation1}
521 \centering
522 {\footnotesize
523 \begin{tabular}{ccc}
524 \toprule
525 Tipo da Publicação & \verb|\cite| & \verb|\citet|\\
526 \midrule
527 Livro & \cite{book-example} & \citet{book-example}\\
528 Artigo & \cite{article-example} & \citet{article-example}\\
529 Relatório & \cite{techreport-example} & \citet{techreport-example}\\
530 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\\
531 Anais de Congresso & \cite{inproceedings-example} &
532 \citet{inproceedings-example}\\
533 Séries & \cite{incollection-example} & \citet{incollection-example}\\
534 Em Livro & \cite{inbook-example} & \citet{inbook-example}\\
535 Dissertação de mestrado & \cite{mastersthesis-example} &
536 \citet{mastersthesis-example}\\
537 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\\
538 \bottomrule

```



```

539 \end{tabular}}
540 \end{table}
541
542 \chapter{Alguns outros exemplo úteis}
543
544 \begin{tcolorbox}[title=Meu Textbox]
545 Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir
546 \end{tcolorbox}
547
548 \begin{tcolorbox}
549 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem como
550 \end{tcolorbox}
551
552 \begin{figure}[ht]
553 \centering
554 \begin{tikzpicture}
555 \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\textwidth]{image.png}}
556 \begin{scope}[x={(image.south east)},y={(image.north west)}]
557 % Definindo o textbox dentro da figura
558 \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text=white]
559 \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox dentro da figura]
560 Este textbox fala sobre como inserir um textbox dentro de uma figura usando tcolorbox
561 \end{tcolorbox}
562 };
563 \end{scope}
564 \end{tikzpicture}
565 \caption{Figura com Textbox}
566 \label{fig:figura_com_textbox1}
567 \end{figure}
568
569
570 \begin{figure}[ht]
571 \centering
572 \begin{tcolorbox}
573 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem como
574 \end{tcolorbox}
575 \caption{Figura com Textbox simples}
576 \label{fig:figura_com_textbox}
577 \end{figure}
578
579 \section{Listagens de código e algoritmos}
580
581 O CoppeTeX traz embutido o suporte a listagens de programas (pacote \verb|listings|) e a algoritmos (pacote \verb|algorithms|).
582
583 \begin{python}[caption={Exemplo de programa em Python},label=prog:exemplo]
584 def saudacao(nome):
585     # imprime uma saudação acentuada
586     print(f"Olá, {nome}!")
587
588 saudacao("COPPE")
589 \end{python}
590 \source{Elaboração própria.}
591
592 Há estilos prontos também para \verb|java|, \verb|xml|, \verb|html| e \verb|prolog|, comando

```

```

593
594 \begin{algorithm}[H]
595 \caption{Soma dos elementos de um vetor}
596 \label{alg:soma}
597 \Dados{vetor  $v$  com  $n$  elementos}
598 \Resultado{soma  $s$  dos elementos de  $v$ }
599  $s \leftarrow 0$ ;
600 \Para{ $i \leftarrow 1$  \Ate{ }  $n$ }{
601    $s \leftarrow s + v[i]$ ;
602 }
603 \Retorna  $s$ ;
604 \end{algorithm}
605
606 \chapter{Método Proposto}
607 \chapter{Resultados e Discussões}
608
609 \section{Algumas Demonstrações}
610
611 A Lista de Símbolos precisa usar comandos específicos. Aqui vamos usar os símbolos  $\alpha$ 
612 \syml[beta]{Beta}{A palavra Beta more e corrigida}
613 \syml[zzbeta]{ $\beta$ }{A letra  $\beta$  corrigida}
614 \syml{beta}{A palavra beta}
615 \syml{alpha}{A palavra alpha}
616 \syml[alpha]{Alpha}{A palavra Alpha}
617 \syml[zzalpha]{ $\alpha$ }{A letra  $\alpha$  corrigida}
618 \syml[marco]{Marco}{A palavra Marco corrigida}
619
620 A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exemplos
621 \abbrev{GoT}{Game of Thrones}
622 \abbrev[GOT]{GoT}{Game of Thrones ordenado como GOT}
623 \abbrev[iot]{IoT}{IoT ordenado como iot}
624 \abbrev[IoT]{IoT}{IoT ordenado como IoT}
625 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
626 \abbrev{IoT}{IoT com ordenação default}
627 \abbrev[ITU]{ITU}{ITU mesmo}
628
629 \newsigla{ufrj}{UFRJ}{Universidade Federal do Rio de Janeiro}
630 \newsigla{cpgp}{CPGP}{Comissão de Pós-Graduação e Pesquisa de Engenharia}
631
632 As siglas (manual2024, seção 2.8) são expandidas na primeira ocorrência e registradas automaticamente
633
634
635
636
637 \chapter{Conclusões}
638
639 \backmatter
640
641 % Tipos exóticos da ABNT (NBR 6023): patente, legislação, norma, mapa,
642 % partitura --- declarados via sinônimos em example.bib.
643 \nocite{patente-example,legislacao-example,norma-example,mapa-example,partitura-example}
644 \nocite{videogame-example,filme-example,semautor-example}
645 \nocite{jornalcaderno-example,jornal-example}
646 \printbibliography

```

```

647
648
649
650 \appendix
651
652 \chapter{Um apêndice}
653
654 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
655
656 Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo
657
658
659 \chapter{Outro apêndice}
660
661 \annex
662
663
664 \chapter{Um Anexo}
665 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
666
667
668
669 Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que
670
671 No modelo \CoppeTeX os anexos devem obrigatoriamente vir depois dos apêndices e usam o comand
672
673
674 \chapter{Outro Anexo}
675
676
677 \end{document}
678
679 \example

```

8 Implementation

8.1 The ‘coppe.cls’ file

```

680 \class
681 \def\filename{coppe.dtx}
682 \def\fileversion{v3.8}
683 \def\filedate{2026/05/25}
684 \NeedsTeXFormat{LaTeX2e}[1995/12/01]
685 \ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
686 \LoadClass[12pt,a4paper,twoside]{book}
687 % The bibliography engine (biblatex+biber) is loaded further below, after babel,
688 % so the numbers class option can select the citation style at load time.
689 \RequirePackage{hyphenat}
690 \RequirePackage{lastpage}
691 \RequirePackage{ifthen}
692 \RequirePackage{graphicx}
693 \RequirePackage{setspace}
694 \RequirePackage{tabularx}
695 \RequirePackage{etoolbox}

```

```

696 \RequirePackage{eqparbox}
697 \RequirePackage{ltxcmds}
698 \RequirePackage{hologo} % TeX-family logos: \hologo{TeX}, \hologo{LaTeX}, ...
699 % Render \TeX, \LaTeX and friends via hologo (\CoppeTeX is a custom logo, kept).
700 \def\TeX{\hologo{TeX}}
701 \def\LaTeX{\hologo{LaTeX}}
702 \def\LaTeXe{\hologo{LaTeXe}}
703 \def\BibTeX{\hologo{BibTeX}}
704 \def\pdfLaTeX{\hologo{pdfLaTeX}}
705 \def\pdfTeX{\hologo{pdfTeX}}
706 \def\LuaLaTeX{\hologo{LuaLaTeX}}
707 \def\XeLaTeX{\hologo{XeLaTeX}}
708 \def\MiKTeX{\hologo{MiKTeX}}
709 \RequirePackage[T1]{fontenc}
710 % manual2024 margins: 3cm top, 2cm bottom, mirrored 3cm inner / 2cm outer for
711 % two-sided printing -- realised as inner=outer=2cm plus a 1cm bindingoffset.
712 \RequirePackage[a4paper,twoside,bindingoffset=1cm,%
713 top=3cm,bottom=2cm,inner=2cm,outer=2cm]{geometry}
714 \RequirePackage{titlesec}
715 % --- Section heading formats (manual2024 2.6 / proposta P22) -----
716 % chapter (numbered): "N TITLE", ALL CAPS, bold, flush left (no "Capitulo N").
717 % APENDICE/ANEXO chapters print "APENDICE A -- Title" (R75/R76); normal chapters
718 % keep "N Title". \appendix and \annex raise the flag (see below).
719 \newif\if@coppeappendix \@coppeappendixfalse
720 \titleformat{chapter}[hang]
721 {\normalfont\Large\bfseries}
722 {\if@coppeappendix\MakeUppercase{\appendixname}\ \thechapter\ \textendash\else\thechapter\
723 {1ex}}{\MakeUppercase}
724 \titlespacing*{chapter}{0pt}{10pt}{1.5\baselineskip}
725 \let\coppe@orig@appendix\appendix
726 \renewcommand\appendix{\coppe@orig@appendix\@coppeappendixtrue}
727 % Sumario (TOC) entries for APENDICE/ANEXO chapters read "APENDICE A -- Title"
728 % / "ANEXO A -- Title" (R75/R76), not the bare "A". The label word is chosen by
729 % which macro is written (so it is correct at .toc-render time even though the
730 % appendixname has changed back); the chapter letter is baked in at write time.
731 \newif\if@coppeannex \@coppeannexfalse
732 \newcommand\coppe@tocapp[1]{%
733 \MakeUppercase{\iflanguage{brazilian}{Ap\^endice}{Appendix}}}%
734 \nobreakspace#1\nobreakspace\textendash\nobreakspace}
735 \newcommand\coppe@tocanx[1]{%
736 \MakeUppercase{\iflanguage{brazilian}{Anexo}{Annex}}}%
737 \nobreakspace#1\nobreakspace\textendash\nobreakspace}
738 \def\@chapter[#1]#2{%
739 \ifnum \c@secnumdepth >\m@ne
740 \if@mainmatter
741 \refstepcounter{chapter}%
742 \typeout{\@chapapp\space\thechapter.}%
743 \if@coppeappendix
744 \if@coppeannex
745 \addcontentsline{toc}{chapter}{\protect\coppe@tocanx{\thechapter}#1}%
746 \else
747 \addcontentsline{toc}{chapter}{\protect\coppe@tocapp{\thechapter}#1}%
748 \fi
749 \else

```

```

750      \addcontentsline{toc}{chapter}{\protect\numberline{\thechapter}\#1}%
751      \fi
752      \else
753      \addcontentsline{toc}{chapter}{\#1}%
754      \fi
755      \else
756      \addcontentsline{toc}{chapter}{\#1}%
757      \fi
758      \chaptermark{\#1}%
759      \addtocontents{lof}{\protect\addvspace{10\p@}}%
760      \addtocontents{lot}{\protect\addvspace{10\p@}}%
761      \if@twocolumn
762      \@topnewpage[\@makechapterhead{\#2}]%
763      \else
764      \@makechapterhead{\#2}%
765      \@afterheading
766      \fi}
767 % chapter (unnumbered): centred ALL CAPS bold (errata, lists, resumo, sumario,
768 % referencias, glossario, ... -- unnumbered headings are centred, P29-P30).
769 \titleformat{name=\chapter,numberless}[hang]
770   {\normalfont\Large\bfseries\filcenter}{\#1}{0pt}{\MakeUppercase}
771 \titlespacing*{name=\chapter,numberless}{0pt}{10pt}{1.5\baselineskip}
772 % section: ALL CAPS, normal weight.
773 \titleformat{section}
774   {\normalfont\large\bfseries}{\thesection}{1ex}{\MakeUppercase}
775 % subsection: bold (author writes Title Case).
776 \titleformat{subsection}
777   {\normalfont\bfseries}{\thesubsection}{1ex}{\bfseries}
778 % subsubsection: bold italic.
779 \titleformat{subsubsection}
780   {\normalfont\bfseries\itshape}{\thesubsubsection}{1ex}{\bfseries}
781 % --- Captions (manual2024 2.10-2.11 / proposta P6): smaller font, ABNT
782 % "Figura 1 -- titulo" travessao separator (sources handled by \source). ---
783 \RequirePackage{caption}
784 \captionsetup{font=footnotesize,labelsep=endash}
785 % --- Floats: caption on TOP and a mandatory "Fonte:" line at the bottom.
786 % manual2024 2.10-2.11 (R46/R51: identification on top; R48/R52: the source
787 % at the bottom is mandatory). The float package's plaintop style forces the
788 % caption above the body for figure, table and the new quadro float. -----
789 \RequirePackage{float}
790 \floatstyle{plaintop}
791 \restylefloat{figure}
792 \restylefloat{table}
793 % New "quadro" float -- "Quadro" (pt) / "Frame" (en) (R53) -- with its own list
794 % file (.loq). A quadro's data is bordered on all sides (drawn by the author),
795 % unlike a tabela (rules on top and bottom only).
796 \providecommand{\quadroname}{\iflanguage{brazilian}{Quadro}{Frame}}
797 \providecommand{\listquadroname}{%
798   \iflanguage{brazilian}{Lista de Quadros}{List of Frames}}
799 \newfloat{quadro}{htbp}{loq}[chapter]
800 \floatname{quadro}{\quadroname}
801 \providecommand{\quadroautorefname}{\quadroname}
802 % Format the Lista de Quadros entries exactly like the Lista de Figuras ones
803 % (float's own \l@quadro lays the number/title out incorrectly).

```

```

804 \let\l@quadro\l@figure
805 \newcommand\listofquadros{%
806   \chapter*{\listquadroname}%
807   \addcontentsline{toc}{chapter}{\listquadroname}%
808   \@mkboth{\MakeUppercase\listquadroname}{\MakeUppercase\listquadroname}%
809   \@starttoc{loq}}
810 % \source / \fonte: the illustration source (ABNT-mandatory). Typeset below the
811 % float as a smaller, single-spaced, centred line; it does NOT advance any
812 % caption counter. \cpsource/\cpfonte are clash-safe aliases (the rename rule);
813 % the label is "Fonte" (pt) / "Source" (en).
814 \newcommand{\cpsourcenam}{\iflanguage{brazilian}{Fonte}{Source}}
815 \newcommand{\cpsource}[1]{%
816   \par\addvspace{\smallskipamount}%
817   {\footnotesize\singlespacing\centering\cpsourcenam: #1\par}}
818 \let\cpfonte\cpsource
819 \@ifundefined{source}{\let\source\cpsource}{}%
820 \@ifundefined{fonte}{\let\fonte\cpsource}{}%
821 % \newcoppefloat{env}{caption name}{list title}: declare an extra caption-on-top
822 % float (e.g. "picture") that supports \source and gains a \listof<env>s command.
823 \newcommand{\newcoppefloat}[3]{%
824   {\floatstyle{plaintop}\newfloat{#1}{htbp}{lo#1}}%
825   \floatname{#1}{#2}%
826   \expandafter\newcommand\csname listof#1\endcsname{%
827     \chapter*{#3}\addcontentsline{toc}{chapter}{#3}%
828     \@mkboth{\MakeUppercase{#3}}{\MakeUppercase{#3}}%
829     \@starttoc{lo#1}}}
830 % --- Footnotes (manual2024 / R16, R8, R21): single spacing and a 5 cm rule
831 % (filete) from the left margin; footnotesize is the book default. -----
832 \let\coppe@orig@footnotetext\@footnotetext
833 \long\def\@footnotetext#1{\coppe@orig@footnotetext{\singlespacing#1}}
834 \renewcommand\footnoterule{\kern-3\p@\hrule\@width 5cm\kern 2.6\p@}
835 % --- alineas: list keyed by lowercase letters a) b) c); subitems with "--"
836 % (manual2024 2.6 / proposta P24). -----
837 \RequirePackage{enumitem}
838 \newlist{alineas}{enumerate}{2}
839 \setlist{alineas,1}{label=\alph*},leftmargin=1.8em,itemsep=0pt}
840 \setlist{alineas,2}{label=--,leftmargin=1.8em,itemsep=0pt}
841 % --- Page numbering: arabic in the top OUTER corner on textual pages
842 % (manual2024 2.7 / proposta P25-P26). Pre-textual numbering is set by
843 % \frontmatter (unnumbered by default; roman with the numeraisromanos
844 % class option). -----
845 \RequirePackage{fancyhdr}
846 \fancypagestyle{coppe}{\fancyhf{}\fancyhead[R0,LE]{\thepage}%
847 \renewcommand{\headrulewidth}{0pt}}
848 % Blank pages inserted by \cleardoublepage (two-sided) must be empty: no page
849 % number may leak onto them (e.g. the verso right after the cover).
850 \renewcommand{\cleardoublepage}{%
851   \clearpage
852   \if@twoside
853     \ifodd\c@page\else
854       \hbox{}\thispagestyle{empty}\newpage
855       \if@twocolumn\hbox{}\newpage\fi
856     \fi
857   \fi}

```

```

858 \def\CoppeTeX{{\rm C\kern-.05em{\sc o\kern-.025em p\kern-.025em
859 p\kern-.025em e}}\kern-.08em
860 T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}

861 \newboolean{maledoc}
862 \setboolean{maledoc}{false}
863 %
864 % Class options.
865 % If you are writing a text in English, you must turn ‘‘English’’ on.
866 % Otherwise, Portuguese is considered the main language.
867 \newif\if@english\@englishfalse
868 \DeclareOption{english}{\@englishtrue}
869 \DeclareOption{msc}{%
870   \newcommand{\@degree}{M.Sc.}
871   \newcommand{\@degree name}{Mestrado}
872   \newcommand{\local@degname}{Mestre}
873   \newcommand{\foreign@degname}{Master}
874   \newcommand\local@doctype{Disserta{\c c}{\~ a}o}
875   \newcommand\foreign@doctype{Dissertation}
876 }
877 \DeclareOption{dscexam}{%
878   \newcommand{\@degree}{D.Sc.}
879   \newcommand{\@degree name}{Doutorado}
880   \newcommand{\local@degname}{Doutor}
881   \newcommand{\foreign@degname}{Doctor}
882   \setboolean{maledoc}{true}
883   \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
884   \newcommand\foreign@doctype{Qualifying Exam}
885 }
886 \DeclareOption{mscexam}{%
887   \newcommand{\@degree}{M.Sc.}
888   \newcommand{\@degree name}{Mestrado}
889   \newcommand{\local@degname}{Mestre}
890   \newcommand{\foreign@degname}{Master}
891   \setboolean{maledoc}{true}
892   \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
893   \newcommand\foreign@doctype{Qualifying Exam}
894 }
895 \DeclareOption{dsc}{%
896   \newcommand{\@degree}{D.Sc.}
897   \newcommand{\@degree name}{Doutorado}
898   \newcommand{\local@degname}{Doutor}
899   \newcommand{\foreign@degname}{Doctor}
900   \newcommand\local@doctype{Tese}
901   \newcommand\foreign@doctype{Thesis}
902 }
903 \newif\if@coppenumeric\@coppenumericfalse
904 \DeclareOption{numbers}{\@coppenumerictrue}
905 % numeraisromanos: show roman numerals on the pre-textual pages. Default: the
906 % pre-textual pages are counted but not numbered (manual2024 2.7).
907 \newif\if@copperoman\@copperomanfalse
908 \DeclareOption{numeraisromanos}{\@copperomantrue}

```

Here is the default one-and-a-half line spacing. Users can change to double spacing by passing the `doublespacing` option.

```

909 \onehalfspacing
910 \DeclareOption{doublespacing}{%
911   \doublespacing
912 }

913 \ProcessOptions\relax
914 \if@english
915   \RequirePackage[brazilian,english]{babel}
916 \else
917   \RequirePackage[english,brazilian]{babel}
918 \fi

919 % Quotation marks: csquotes gives the language-aware \enquote; \citacao is its
920 % Portuguese alias for short, inline quotations (proposta P3).
921 \RequirePackage{csquotes}
922 \newcommand{\citacao}[1]{\enquote{#1}}
923 % --- Code listings & algorithms (TOD03 "tratar listagens"): the listings
924 %   package gives the "Programa" listing (caption on top, ABNT R46) with
925 %   \listofprogramas; algorithm2e gives algorithms. Names are babel-aware and
926 %   the language styles below are based on the user's listagens.tex. -----
927 \RequirePackage{xcolor}
928 \RequirePackage{listings}
929 \RequirePackage{fancyvrb}
930 \definecolor{deepblue}{rgb}{0,0,0.5}
931 \definecolor{deepred}{rgb}{0.6,0,0}
932 \definecolor{deepgreen}{rgb}{0,0.5,0}
933 \DeclareFixedFont{\ttb}{T1}{txtt}{bx}{n}{10}
934 \DeclareFixedFont{\ttm}{T1}{txtt}{m}{n}{10}
935 \newcommand{\gxnumbersep}{0.5em}
936 \lstset{captionpos=t,extendedchars=true,basicstyle=\ttfamily,inputencoding=utf8,%
937   literate=%
938   {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{e}}1%
939   {\~}{\~{e}}1 {\~}{\~{e}}1 {\~}{\~{i}}1 {\~}{\~{i}}1 {\~}{\~{o}}1%
940   {\~}{\~{o}}1 {\~}{\~{o}}1 {\~}{\~{u}}1 {\~}{\~{u}}1 {\~}{\~{c}}1%
941   {\~}{\~{C}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{E}}1%
942   {\~}{\~{E}}1 {\~}{\~{I}}1 {\~}{\~{I}}1 {\~}{\~{O}}1 {\~}{\~{O}}1 {\~}{\~{U}}1%
943   {\~}{\~{U}}1}
944 }
945 % Babel-aware listing names, set after listings' own babel setup.
946 \AtBeginDocument{%
947   \renewcommand{\lstlistingname}{\iflanguage{brazilian}{Programa}{Program}}%
948   \renewcommand{\lstlistlistingname}{%
949     \iflanguage{brazilian}{Lista de Programas}{List of Programs}}%
950 % \listofprogramas in the coppe list style (heading + ToC + running head).
951 \newcommand{\listofprogramas}{%
952   \chapter*{\lstlistlistingname}%
953   \addcontentsline{toc}{chapter}{\lstlistlistingname}%
954   \@mkboth{\MakeUppercase\lstlistlistingname}{\MakeUppercase\lstlistlistingname}%
955   \@starttoc{lol}}
956 % Per-language styles + environments (python, java, xml, html, prolog).
957 \newcommand{\pythonstyle}{\lstset{language=Python,basicstyle=\ttm,numbers=left,%
958   numberstyle=\small\ttfamily,numbersep=\gxnumbersep,commentstyle=\color{purple},%
959   morekeywords={self,as},keywordstyle=\ttb\color{deepblue},%
960   emph={MyClass,__init__},emphstyle=\ttb\color{deepred},%
961   stringstyle=\color{deepgreen},frame=tb,showstringspaces=false,%
962   extendedchars=true,firstnumber=auto,inputencoding=utf8,breaklines=true,%

```



```

963 escapechar=' ,postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
964 \lstnewenvironment{python}[1][\pythonstyle\lstset{#1}%
965 \csname\@lst @SetFirstNumber\endcsname\csname\@lst @SaveFirstNumber\endcsname}
966 \newcommand\pythoninline[1][\pythonstyle\lstinline!#1!}]
967 \newcommand\pythonexternal[2][\pythonstyle\lstinputlisting[1]{#2}]}
968 \newcommand\xmlstyle{\lstset{language=XML,basicstyle=\ttm,numbers=left,%
969 numberstyle=\small\ttfamily,numbersep=\gxnumbersep,%
970 morekeywords={xs:schema,xs:element,xs:sequence,xs:complexType},%
971 keywordstyle=\ttb\color{deepblue},%
972 emph={name,type,attributeFormDefault,elementFormDefault,xmlns:xs},%
973 emphstyle=\ttb\color{deepred},stringstyle=\color{deepgreen},frame=tb,%
974 showstringspaces=false,extendedchars=true,inputencoding=utf8,breaklines=true,%
975 postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
976 \lstnewenvironment{xml}[1][\xmlstyle\lstset{#1}]{%
977 \newcommand\htmlstyle{\lstset{language=HTML,basicstyle=\ttm,numbers=left,%
978 numberstyle=\small\ttfamily,numbersep=\gxnumbersep,%
979 keywordstyle=\ttb\color{deepblue},emphstyle=\ttb\color{deepred},%
980 stringstyle=\color{deepgreen},frame=tb,showstringspaces=false,%
981 extendedchars=true,inputencoding=utf8,breaklines=true,%
982 postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
983 \lstnewenvironment{html}[1][\htmlstyle\lstset{#1}]{%
984 \newcommand\prologstyle{\lstset{language=Prolog,basicstyle=\ttm,numbers=left,%
985 numberstyle=\small\ttfamily,numbersep=\gxnumbersep,commentstyle=\color{purple},%
986 keywordstyle=\ttb\color{deepblue},emph={MyClass,__init__},%
987 emphstyle=\ttb\color{deepred},stringstyle=\color{deepgreen},frame=tb,%
988 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
989 breaklines=true,escapechar=',%
990 postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
991 \lstnewenvironment{prolog}[1][\prologstyle\lstset{#1}%
992 \csname\@lst @SetFirstNumber\endcsname\csname\@lst @SaveFirstNumber\endcsname}
993 \newcommand\javastyle{\lstset{language=Java,basicstyle=\ttm,numbers=left,%
994 numberstyle=\small\ttfamily,numbersep=\gxnumbersep,commentstyle=\color{purple},%
995 morekeywords={self,as},keywordstyle=\ttb\color{deepblue},%
996 emphstyle=\ttb\color{deepred},stringstyle=\color{deepgreen},frame=tb,%
997 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
998 breaklines=true,escapechar=',%
999 postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
1000 \lstnewenvironment{java}[1][\javastyle\lstset{#1}%
1001 \csname\@lst @SetFirstNumber\endcsname\csname\@lst @SaveFirstNumber\endcsname}
1002 \newcommand\javainline[1][\javastyle\lstinline!#1!}]
1003 \newcommand\javaexternal[2][\javastyle\lstinputlisting[1]{#2}]}
1004 % Verbatim output environments (program output), fancyvrb.
1005 \DefineVerbatimEnvironment{gxoutput}{Verbatim}%
1006 {frame=lines,numbers=left,numbersep=\gxnumbersep}
1007 \DefineVerbatimEnvironment{gxoutputs}{Verbatim}%
1008 {frame=lines,numbers=left,numbersep=\gxnumbersep,fontsize=\small}
1009 \renewcommand{\theFancyVerbLine}{\small\ttfamily\arabic{FancyVerbLine}}
1010 % Algorithms: algorithm2e, language following the english class option.
1011 \if@english
1012 \RequirePackage[linesnumbered,lined,algochapter,ruled]{algorithm2e}
1013 \else
1014 \RequirePackage[portuguese,linesnumbered,lined,algochapter,ruled]{algorithm2e}
1015 \fi
1016 \SetKwFor{Para}{para}{faça}{fim do para}

```

```

1017 \SetKwFor{Enquanto}{enquanto}{faça}{fim do enquanto}
1018 % Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
1019 % former natbib+bibtex setup. Loaded here, after babel, so its language strings
1020 % map correctly. The numbers option selects the numeric style; otherwise the
1021 % author-date default is used. natbib=true keeps \citel/\citep available.
1022 \if@coppenumeric
1023   \RequirePackage[backend=biber,datamodel=coppe,%
1024     bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{biblatex}
1025 \else
1026   \RequirePackage[backend=biber,datamodel=coppe,%
1027     bibstyle=coppe,citestyle=coppe,natbib=true]{biblatex}
1028 \fi
1029 % Reference-list heading "Referencias" (pt) / "References" (en) -- manual2024
1030 % 3.1.4.1 requires the post-textual list to be titled "Referencias", not
1031 % "Bibliografia" (babel's default \bibname). The title is set directly with
1032 % \iflanguage so it is correct in BOTH the chapter heading and the .toc entry:
1033 % \iflanguage resolves to plain text and survives the write to the .toc, whereas
1034 % biblatex's \bibstring does not, and babel's \bibname caption override proved
1035 % unreliable here. Evaluated in the body, the current language is settled.
1036 \defbibheading{bibliography}{%
1037   \chapter*{\iflanguage{brazilian}{Refer\~encias}{References}}%
1038   \addcontentsline{toc}{chapter}{\iflanguage{brazilian}{Refer\~encias}{References}}}
1039 % The catalog page range (bib:begin/bib:end) was written by the old
1040 % thebibliography redefinition, which biblatex's \printbibliography does not
1041 % use. biblatex offers \AtBeginBibliography but no end counterpart, so the
1042 % begin label uses that hook and the end label is emitted by wrapping
1043 % \printbibliography to write it just after the reference list.
1044 \AtBeginBibliography{\coppe@bibBegin}
1045 \let\coppe@orig@printbibliography\printbibliography
1046 \renewcommand\printbibliography[1][]{%
1047   \coppe@orig@printbibliography[#1]%
1048   \coppe@bibEnd}

```

\department This macro is used to set the author's affiliation. There are twelve options which correspond to all academic units at COPPE/UFRJ. It defines the current and the foreign names of these units.

```

1049 \newcommand\department[1]{%
1050   \ifthenelse{\equal{#1}{PEB}}{
1051     {\global\def\local@deptname{Engenharia Biom(\` e)dica}
1052     \global\def\foreign@deptname{Biomedical Engineering}}{
1053     \ifthenelse{\equal{#1}{PEC}}{
1054       {\global\def\local@deptname{Engenharia Civil}
1055       \global\def\foreign@deptname{Civil Engineering}}{
1056       \ifthenelse{\equal{#1}{PEE}}{
1057         {\global\def\local@deptname{Engenharia El(\` e)trica}
1058         \global\def\foreign@deptname{Electrical Engineering}}{
1059         \ifthenelse{\equal{#1}{PEM}}{
1060           {\global\def\local@deptname{Engenharia Mec(\` a)nica}
1061           \global\def\foreign@deptname{Mechanical Engineering}}{
1062           \ifthenelse{\equal{#1}{PEMM}}{
1063             {\global\def\local@deptname{Engenharia Metal(\` u)rgica e de Materiais}
1064             \global\def\foreign@deptname{Metallurgical and Materials Engineering}}{
1065             \ifthenelse{\equal{#1}{PEN}}{
1066               {\global\def\local@deptname{Engenharia Nuclear}

```

```

1067 \global\def\foreign@deptname{Nuclear Engineering}}{}
1068 \ifthenelse{equal{#1}{PEN0}}{
1069   {\global\def\local@deptname{Engenharia Oce{\~ a}nica}
1070   \global\def\foreign@deptname{Ocean Engineering}}{}
1071 \ifthenelse{equal{#1}{PPE}}{
1072   {\global\def\local@deptname{Planejamento Energ{\~ e}tico}
1073   \global\def\foreign@deptname{Energy Planning}}{}
1074 \ifthenelse{equal{#1}{PEP}}{
1075   {\global\def\local@deptname{Engenharia de Produ{\c c}{\~ a}o}
1076   \global\def\foreign@deptname{Production Engineering}}{}
1077 \ifthenelse{equal{#1}{PEQ}}{
1078   {\global\def\local@deptname{Engenharia Qu{\~ i}mica}
1079   \global\def\foreign@deptname{Chemical Engineering}}{}
1080 \ifthenelse{equal{#1}{PESC}}{
1081   {\global\def\local@deptname{Engenharia de Sistemas e Computa{\c c}{\~ a}o}
1082   \global\def\foreign@deptname{Systems Engineering and Computer Science}}{}
1083 \ifthenelse{equal{#1}{PET}}{
1084   {\global\def\local@deptname{Engenharia de Transportes}
1085   \global\def\foreign@deptname{Transportation Engineering}}{}
1086 \ifthenelse{equal{#1}{PENT}}{
1087   {\global\def\local@deptname{Engenharia de Nanotecnologia}
1088   \global\def\foreign@deptname{Nanotechnology Engineering}}{}
1089 }

\title Used to enter the title in Brazilian Portuguese.
1090 \renewcommand\title[1]{%
1091 \global\def\local@title{#1}%
1092 }

\foreigntitle Used to enter the foreign title.
1093 \newcommand\foreigntitle[1]{%
1094 \global\def\foreign@title{#1}%
1095 }

\advisor Defines globally the title, name and academic degree of the advisors.
1096 \newcount\@advisor\@advisor0
1097 \newcommand\advisor[4]{%
1098 \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#1}
1099 \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#2}
1100 \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#3}
1101 \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#4}
1102 \global\advance\@advisor by 1
1103 \ifnum\@advisor>1
1104 \renewcommand\local@advisorstring{Orientadores}
1105 \renewcommand\foreign@advisorstring{Advisors}
1106 \fi
1107 }

\examiner
1108 \newcount\@examiner\@examiner0
1109 \newcommand\examiner[3]{%
1110 \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#1\ #2}
1111 \global\advance\@examiner by 1
1112 }

```

`\author` It was redefined to allow the identification of the author's first names and surname.

```
1113 \renewcommand\author[2]{%
1114   \global\def\@authname{#1}
1115   \global\def\@authsurn{#2}
1116 }
```

`\date` This code makes easy to switch from dates in different languages.

```
1117 \renewcommand\date[2]{%
1118   \month=#1
1119   \year=#2
1120 }
```

`\local@monthname`

```
1121 \newcommand\local@monthname{\ifcase\month\or
1122   Janeiro\or Fevereiro\or Mar{\c c}\o\or Abril\or Maio\or Junho\or
1123   Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}
```

`\foreign@monthname`

```
1124 \newcommand\foreign@monthname{\ifcase\month\or
1125   January\or February\or March\or April\or May\or June\or
1126   July\or August\or September\or October\or November\or December\fi}
```

`\keyword`

```
1127 \newcounter{keywords}
1128 \newcommand\keyword[1]{%
1129   \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}
1130   \global\addtocounter{keywords}{1}
1131 }
```

`\freeconfig` This command allows easy changing of core class parameters.

```
1132 \newcommand\freeconfig[6]{%
1133   \providecommand\@degree{}
1134   \renewcommand\@degree{#1}
1135   \providecommand\@degreename{}
1136   \renewcommand\@degreename{#2}
1137   \providecommand\local@degname{}
1138   \renewcommand\local@degname{#3}
1139   \providecommand\foreign@degname{}
1140   \renewcommand\foreign@degname{#4}
1141   \providecommand\local@doctype{}
1142   \renewcommand\local@doctype{#5}
1143   \providecommand\foreign@doctype{}
1144   \renewcommand\foreign@doctype{#6}%
1145 }
1146 % \end{macrocode}
1147 % \end{macro}
1148 %
1149 % \begin{macro}{\frontmatter}
1150 % The number of pages for both frontmatter and mainmatter printed
1151 % in the cataloging details page is computed by means of simple
1152 % \LaTeX\ labels.
1153 % \begin{macrocode}
1154 \renewcommand\frontmatter{%
```

```

1155 \cleardoublepage
1156 \@mainmatterfalse
1157 \pagenumbering{roman}
1158 \thispagestyle{empty}
1159 % folha de rosto = counted page 1 (ABNT: the capa is not counted). An odd
1160 % start keeps the page-counter parity aligned with the physical (recto) side,
1161 % so \cleardoublepage lands every pre-textual element on an odd (front) page.
1162 \setcounter{page}{1}
1163 \makefrontpage
1164 \clearpage
1165 \if@copperoman
1166   \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\thepage}}%
1167   \renewcommand{\headrulewidth}{0pt}}%
1168   \pagestyle{coppe}%
1169 \else
1170   \fancypagestyle{plain}{\fancyhf{}\renewcommand{\headrulewidth}{0pt}}%
1171   \pagestyle{empty}%
1172 \fi
1173 }

\mainmatter
1174 \renewcommand\mainmatter{%
1175   \coppe@mainBegin
1176   \cleardoublepage
1177   \@mainmattertrue
1178   \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\thepage}}%
1179   \renewcommand{\headrulewidth}{0pt}}%
1180   \pagestyle{coppe}%
1181   \pagenumbering{arabic}}

\backmatter
1182 \renewcommand\backmatter{%
1183   \if@openright
1184     \cleardoublepage
1185   \else
1186     \clearpage
1187   \fi}
1188 %

\makecover The ABNT cover (capa, NBR 14724) suggested in the manual's Anexo A. It is
the institution block — the university, the full COPPE name, and the graduate
program taken from \department — followed by the author, the title, and the
city and year, all centred and uppercased. The optional UFRJ and COPPE logos
head the page. The capa is not counted: it precedes the (roman) front matter and
is produced before the titlepage (folha de rosto).
1189 \newcommand\makecover{%
1190   \begin{titlepage}%
1191   \centering
1192   \makebox[\linewidth]{%
1193     \includegraphics[height=2cm]{ufrj-logo}\hfill
1194     \includegraphics[height=2cm]{coppe-logo}}\par
1195   \vspace{1.5cm}
1196   {\singlespacing

```

```

1197 \MakeUppercase{\local@universityname}\par
1198 \nohyphens{\MakeUppercase{Instituto Alberto Luiz Coimbra de
1199 P{\ ' o}s-Gradua{\c c}{\~ a}o e Pesquisa de Engenharia (COPPE)}}\par
1200 \MakeUppercase{Programa de \local@deptname}\par}
1201 \vspace{\stretch{2}}
1202 \textbf{\MakeUppercase{\@authname\ \@authsur}}\par
1203 \vspace{\stretch{2}}
1204 \nohyphens{%
1205 \if@english
1206 \MakeUppercase\foreign@title
1207 \else
1208 \MakeUppercase\local@title
1209 \fi}\par
1210 \vspace{\stretch{3}}
1211 \MakeUppercase{\local@cityname}\par
1212 \number\year
1213 \end{titlepage}%
1214 }

```

\maketitle

```

1215 \renewcommand\maketitle{%
1216 \pagenumbering{alph}
1217 \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}%
1218 \makecover
1219 \begin{titlepage}
1220 \begin{flushleft}
1221 \vspace*{1.5mm}
1222 \setlength\baselineskip{0pt}
1223 \setlength\parskip{1mm}
1224 \makebox[\linewidth]{%
1225 \includegraphics[height=2cm]{ufrj-logo}\hfill
1226 \includegraphics[height=2cm]{coppe-logo}}
1227 \end{flushleft}
1228 \vspace{1.05cm}
1229 \begin{center}
1230 \nohyphens{%
1231 \if@english
1232 \MakeUppercase\foreign@title
1233 \else
1234 \MakeUppercase\local@title
1235 \fi}\par
1236 \vspace*{3cm}
1237 \nohyphens{\@authname\ \@authsur}\par
1238 \end{center}
1239 \vspace*{2.1cm}
1240 \begin{flushright}
1241 \begin{minipage}{8.45cm}
1242 \frontcover@maintext
1243 \end{minipage}\par
1244 \vspace*{7.5mm}
1245 \nohyphens{%
1246 \begin{tabularx}{8.45cm}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}}
1247 \local@advisorstring: &
1248 \count1=0

```

```

1249 \toks@={}
1250 \@whilenum \count1<\@advisor \do{%
1251 \ifcase\count1 % same as \ifnum0=\count1
1252 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1253 \expandafter\endcsname\expandafter\space%
1254 \csname CoppeAdvisorSurname:\the\count1\endcsname\}}
1255 \else
1256 \toks@=\expandafter\expandafter\expandafter{%
1257 \expandafter\the\expandafter\toks@%
1258 \expandafter&\expandafter\space%
1259 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1260 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\}
1261 }%
1262 \fi
1263 \advance\count1 by 1}
1264 \the\toks@
1265 \end{tabularx}}\par
1266 \end{flushright}
1267 \vspace*{\fill}
1268 \begin{center}
1269 \local@cityname\par
1270 \local@monthname\ de \number\year
1271 \end{center}
1272 \end{titlepage}
1273 % Ficha catalográfica on the verso of the folha de rosto (NBR 14724, R57):
1274 % the title page is a recto, so this lands on the immediately following verso,
1275 % ahead of the approval sheet produced by \frontmatter.
1276 \ifthenelse{\boolean{maledoc}}{\thispagestyle{empty}\makecatalog}%
1277 \global\let\maketitle\relax%
1278 \global\let\and\relax}

1279 \newcommand\makefrontpage{%
1280 \begin{center}
1281 \sloppy\nohyphens{
1282 \if@english
1283 \MakeUppercase\foreign@title
1284 \else
1285 \MakeUppercase\local@title
1286 \fi}\par
1287 \vspace*{7mm}
1288 {\@authname\ \@authsurn}\par
1289 \end{center}\par
1290 \vspace*{4mm}
1291 \frontpage@maintext
1292 \vspace*{16mm}
1293 \nohyphens{%
1294 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}
1295 \local@advisorstring: &
1296 \count1=0
1297 \toks@={}
1298 \@whilenum \count1<\@advisor \do{%
1299 \ifcase\count1 % same as \ifnum0=\count1
1300 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1301 \expandafter\endcsname\expandafter\space%

```

```

1302         \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1303     \else
1304         \toks@=\expandafter\expandafter\expandafter{%
1305             \expandafter\the\expandafter\toks@%
1306             \expandafter&\expandafter\space%
1307             \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1308             \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
1309         }%
1310     \fi
1311     \advance\count1 by 1}
1312     \the\toks@
1313 \end{tabularx}\par
1314 \vspace*{20mm}
1315 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}}
1316     Aprovada por: &
1317     \count1=0
1318     \toks@={}
1319     \@whilenum \count1<\@examiner \do{%
1320     \ifcase\count1 % same as \ifnum0=\count1
1321         \toks@=\expandafter{\csname CoppeExaminer:\the\count1%
1322             \expandafter\endcsname\expandafter\\}
1323     \else
1324         \toks@=\expandafter\expandafter\expandafter{%
1325             \expandafter\the\expandafter\toks@%
1326             \expandafter&\expandafter\space%
1327             \csname CoppeExaminer:\the\count1\expandafter\endcsname%
1328             \expandafter\space\\
1329         }%
1330     \fi
1331     \advance\count1 by 1}
1332     \the\toks@
1333 \end{tabularx}}\par
1334 \vspace*{\fill}
1335 \frontpage@bottomtext}

1336 \newcommand\coppe@hypersetup{%
1337 \begingroup
1338 % changes to \toks@ and \count@ are kept local;
1339 % it's not necessary for them, but it is usually the case
1340 % for \count1, because the first ten counters are written
1341 % to the DVI file, thus you got lucky because of PDF output
1342 \toks@={} % in this special case not necessary
1343 \count@=0 %
1344 \@whilenum\count@<\value{keywords}\do{%
1345     % * a keyword separator is not necessary,
1346     %     if there is just one keyword
1347     % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
1348     %     at least once, to get rid of the loop depended \count@
1349     \ifcase\count@ % same as \ifnum0=\count@
1350         \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
1351     \else
1352         \toks@=\expandafter\expandafter\expandafter{%
1353             \expandafter\the\expandafter\toks@
1354             \expandafter;\expandafter\space

```



```

1355         \csname CoppeKeyword:\the\count@\endcsname
1356     }%
1357     \fi
1358     \advance\count@ by 1 %
1359 }%
1360 \edef\x{\endgroup
1361     \noexpand\hypersetup{%
1362         pdfkeywords={\the\toks@}%
1363     }%
1364 }%
1365 \x
1366 \hypersetup{%
1367     pdfauthor={\@authname\ \@authsur},
1368     pdftitle={\local@title},
1369     pdfsubject={\local@doctype\ de \@degree\ em \local@deptname\ da COPPE/UFRJ},
1370     pdfcreator={LaTeX with CoppeTeX toolkit},
1371     breaklinks={true},
1372     raiselinks={true},
1373     pageanchor={true},
1374 }}

```

`\makecatalog` When the document has illustrations, it is required to insert “: il.,” between the number of pages of the textual part and the page dimension. We have created a label to flag the existence of lists of figures. It is checked to be undefined using the plain $\TeX$ command `\isundefined [?]`.

```

1375 \newcommand\makecatalog{%
1376     \vspace*{\fill}
1377     \begin{center}
1378         \setlength{\fboxsep}{5mm}
1379         \framebox[120mm][c]{\makebox[5mm][c]{%
1380             \begin{minipage}[c]{105mm}
1381                 \setlength{\parindent}{5mm}
1382                 \noindent\sloppy\nohyphens\@authsur,
1383                 \nohyphens\@authname\par
1384                 \nohyphens{%
1385                     \if@english
1386                         \foreign@title%
1387                     \else
1388                         \local@title%
1389                     \fi/\@authname\ \@authsur. -- \local@cityname:
1390                     UFRJ/COPPE, \number\year.}\par
1391                     \pageref{front:pageno},
1392                     \pageref{LastPage}
1393                     p.\@ifundefined{r@cat:lofflag}{\pageref{cat:lofflag}} $29,7$cm.\par
1394                     % There is an issue here. When the last entry must be split between lines,
1395                     % the spacing between it and the next paragraph becomes smaller.
1396                     % Should we manually introduce a fixed space? But how could we know that
1397                     % a name was split? Is this happening yet?
1398                 \nohyphens{%
1399                     \begin{tabularx}{100mm}[b]{\@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1400                         \local@advisorstring: &
1401                         \count1=0
1402                         \toks@={}
1403                         \@whilenum \count1<\@advisor \do{%

```

```

1404 \ifcase\count1 % same as \ifnum0=\count1
1405 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1406 \expandafter\endcsname\expandafter\space%
1407 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1408 \else
1409 \toks@=\expandafter\expandafter\expandafter{%
1410 \expandafter\the\expandafter\toks@
1411 \expandafter&\expandafter\space
1412 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1413 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
1414 }%
1415 \fi
1416 \advance\count1 by 1}
1417 \the\toks@
1418 \end{tabularx}}\par
1419 \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
1420 UFRJ/COPPE/Programa de \local@deptname, \number\year.}\par
1421 Refer{\^ e}ncias Bibliogr{\' a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par
1422 \count1=0
1423 \count2=1
1424 \nohyphens{\@whilenum \count1<\value{keywords} \do {%
1425 \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
1426 \advance\count1 by 1
1427 \advance\count2 by 1}
1428 I. \csname CoppeAdvisorSurname:0\endcsname,%
1429 \ \csname CoppeAdvisorName:0\endcsname%
1430 \ifthenelse{\@advisor>1}{\ \emph{et~al.}}{}}.
1431 II. \local@universityname, COPPE, Programa de \local@deptname.
1432 III. T{\' i}tulo.}
1433 \end{minipage}}
1434 \end{center}
1435 \vspace*{\fill}}

```

\dedication

```

1436 \newcommand\dedication[1]{
1437 \gdef\@dedic{#1}
1438 \cleardoublepage
1439 \vspace*{\fill}
1440 \begin{flushright}
1441 \begin{minipage}{60mm}
1442 \raggedleft \it \normalsize \@dedic
1443 \end{minipage}
1444 \end{flushright}}

```

abstract (*env.*) This is a specialization of the abstract in the article standard class.

```

1445 \newenvironment{abstract}{%
1446 \cleardoublepage
1447 \thispagestyle{plain}
1448 \abstract@toptext\par
1449 \vspace*{8.6mm}
1450 \begin{center}
1451 \sloppy\nohyphens{\MakeUppercase\local@title}\par
1452 \vspace*{13.2mm}
1453 \@authname\ \@authsurn \par

```

```

1454 \vspace*{7mm}
1455 \local@monthname/\number\year
1456 \end{center}\par
1457 \vspace*{1.5cm}
1458 \noindent%
1459 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1460 \local@advisorstring: &
1461 \count1=0
1462 \toks@={}
1463 \@whilenum \count1<\@advisor \do{%
1464 \ifcase\count1 % same as \ifnum0=\count1
1465 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1466 \expandafter\endcsname\expandafter\space%
1467 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1468 \else
1469 \toks@=\expandafter\expandafter\expandafter{%
1470 \expandafter\the\expandafter\toks@
1471 \expandafter&\expandafter\space
1472 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1473 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1474 }%
1475 \fi
1476 \advance\count1 by 1}
1477 \the\toks@
1478 \end{tabularx}\par
1479 \vspace*{2mm}
1480 \noindent\local@deptstring: \local@deptname\par
1481 \vspace*{7mm}}{\vspace*{\fill}}

```

foreignabstract (*env.*)

```

1482 \newenvironment{foreignabstract}{%
1483 \cleardoublepage
1484 \thispagestyle{plain}
1485 \begin{otherlanguage}{english}
1486 \foreignabstract@toptext\par
1487 \vspace*{8.6mm}
1488 \begin{center}
1489 \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
1490 \vspace*{13.2mm}
1491 \@authname\ \@authsurn \par
1492 \vspace*{7mm}
1493 \foreign@monthname/\number\year
1494 \end{center}\par
1495 \vspace*{1.5cm}
1496 \noindent%
1497 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1498 \foreign@advisorstring: &
1499 \count1=0
1500 \toks@={}
1501 \@whilenum \count1<\@advisor \do{%
1502 \ifcase\count1 % same as \ifnum0=\count1
1503 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1504 \expandafter\endcsname\expandafter\space%
1505 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}

```

```

1506 \else
1507 \toks@=\expandafter\expandafter\expandafter{%
1508 \expandafter\the\expandafter\toks@
1509 \expandafter&\expandafter\space
1510 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1511 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
1512 }%
1513 \fi
1514 \advance\count1 by 1}
1515 \the\toks@
1516 \end{tabularx}\par
1517 \vspace*{2mm}
1518 \noindent\foreign@deptstring: \foreign@deptname\par
1519 \vspace*{7mm}}{%
1520 \end{otherlanguage}
1521 \vspace*{\fill}
1522 \global\let\@author\@empty
1523 \global\let\@date\@empty
1524 \global\let\foreign@title\@empty
1525 \global\let\foreign@title\relax
1526 \global\let\local@title\@empty
1527 \global\let\local@title\relax
1528 \global\let\author\relax
1529 \global\let\author\relax
1530 \global\let\date\relax}

```

\listoffigures

```

1531 \renewcommand\listoffigures{%
1532 \coppe@hasLof
1533 \if@twocolumn
1534 \@restonecoltrue\onecolumn
1535 \else
1536 \@restonecolfalse
1537 \fi
1538 \chapter*{\listfigurename}%
1539 \addcontentsline{toc}{chapter}{\listfigurename}%
1540 \@mkboth{\MakeUppercase\listfigurename}%
1541 {\MakeUppercase\listfigurename}%
1542 \@starttoc{lof}%
1543 \if@restonecol\twocolumn\fi
1544 }

```

\listoftables

```

1545 \renewcommand\listoftables{%
1546 \if@twocolumn
1547 \@restonecoltrue\onecolumn
1548 \else
1549 \@restonecolfalse
1550 \fi
1551 \chapter*{\listtablename}%
1552 \addcontentsline{toc}{chapter}{\listtablename}%
1553 \@mkboth{%
1554 \MakeUppercase\listtablename}%
1555 {\MakeUppercase\listtablename}%

```

```

1556 \starttoc{lot}%
1557 \if@restonecol\twocolumn\fi
1558 }

```

\printlosymbols

```

1559 \newcommand\printlosymbols{%
1560 \renewcommand\glossaryname{\listsymbolname}%
1561 \@input@{\jobname.los}}

```

\makelosymbols

```

1562 \def\makelosymbols{%
1563 \newwrite\@losfile
1564 \immediate\openout\@losfile=\jobname.syx
1565 \newcommand\syml[3][\@bsphack\begingroup
1566 \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}%
1567 \@sanitize%
1568 \@wrls{\@tempsymb1}{##2}{##3}\typeout%
1569 {Writing index of symbols file \jobname.syx}%
1570 \let\makelosymbols\@empty%
1571 }%
1572 \@onlypreamble\makelosymbols

1573 \AtBeginDocument{%
1574 \ifpackageloaded{hyperref}{%
1575 \newcommand\@wrls[3]{%
1576 \protected@write\@losfile{%
1577 {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
1578 \endgroup%
1579 \@esphack}}{%
1580 \newcommand\@wrls[3]{%
1581 \protected@write\@losfile{%
1582 {\string\indexentry{#1[#2] #3}{\thepage}}%
1583 \endgroup%
1584 \@esphack}}}%

```

\printloabbreviations

```

1585 \newcommand\printloabbreviations{%
1586 \renewcommand\glossaryname{\listabbreviationname}%
1587 \@input@{\jobname.lab}}

```

\makeloabbreviations

```

1588 \def\makeloabbreviations{%
1589 \newwrite\@labfile
1590 \immediate\openout\@labfile=\jobname.abx
1591 \newcommand\abbrev[3][\@bsphack\begingroup
1592 \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}%
1593 \@sanitize%
1594 \@wrlab{\@tempsymb1}{##2}{##3}\typeout%
1595 {Writing index of abbreviations file \jobname.abx}%
1596 \let\makeloabbreviations\@empty
1597 }
1598 \@onlypreamble\makeloabbreviations

```

```

1599 \AtBeginDocument{%
1600 \ifpackageloaded{hyperref}{%
1601   \newcommand\wrlab[3]{%
1602     \protected@write\@labfile{%
1603       {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
1604     \endgroup%
1605     \@esphack}}{%
1606   \newcommand\wrlab[3]{%
1607     \protected@write\@labfile{%
1608       {\string\indexentry{#1[#2] #3}{\thepage}}%
1609     \endgroup%
1610     \@esphack}}}%

\newsigla Acronyms (siglas), manual2024 2.8 / R43. \newsigla{key}{SIGLA}{full form}
\sigla declares one; \sigla{key} prints “full form (SIGLA)” on first use and registers it in
the list of abbreviations and acronyms (through \abbrev, so \makeloabbreviations
must be active), then just “SIGLA” on later uses. \sigla*{key} forces the full
form. The first-use flag is set globally so it survives groups (floats, footnotes).

1611 \newcommand\newsigla[3]{%
1612   \@namedef{coppe@sigla@s#1}{#2}%
1613   \@namedef{coppe@sigla@l#1}{#3}%
1614   \newcommand\coppe@sigla@reg[1]{%
1615     \ifundefined{abbrev}{}%
1616     {\abbrev{\@nameuse{coppe@sigla@s#1}}{\@nameuse{coppe@sigla@l#1}}}%
1617   \newcommand\coppe@sigla@full[1]{%
1618     \@nameuse{coppe@sigla@l#1}\space(\@nameuse{coppe@sigla@s#1})}
1619   \newcommand\coppe@sigla@undef[1]{%
1620     \PackageWarning{coppe}{Sigla '#1' nao declarada com \string\newsigla}%
1621     \textbf{???#1???}}
1622   \newcommand\coppe@sigla@first[1]{%
1623     \global\@namedef{coppe@sigla@done#1}{\coppe@sigla@reg{#1}\coppe@sigla@full{#1}}
1624   \newcommand\coppe@sigla@norm[1]{%
1625     \ifundefined{coppe@sigla@s#1}{\coppe@sigla@undef{#1}}%
1626     {\@ifundefined{coppe@sigla@done#1}{\coppe@sigla@first{#1}}%
1627      {\@nameuse{coppe@sigla@s#1}}}%
1628   \newcommand\coppe@sigla@star[1]{%
1629     \ifundefined{coppe@sigla@s#1}{\coppe@sigla@undef{#1}}%
1630     {\global\@namedef{coppe@sigla@done#1}{\coppe@sigla@full{#1}}}%
1631   \newcommand\sigla{\@ifstar\coppe@sigla@star\coppe@sigla@norm}

1632 %%% \AtBeginDocument{%
1633 %%% \ifpackageloaded{hyperref}{%
1634 %%%   \def\wrlab#1#2{%
1635 %%%     \protected@write\@labfile{%
1636 %%%       {\string\indexentry{[#1] #2|hyperpage}{\thepage}}%
1637 %%%     \endgroup
1638 %%%     \@esphack}}{%
1639 %%%   \def\wrlab#1#2{%
1640 %%%     \protected@write\@labfile{%
1641 %%%       {\string\indexentry{[#1] #2}{\arabic{page}}}%
1642 %%%     \endgroup
1643 %%%     \@esphack}}

1644 % Some macros used to generate cataloging information.
1645 \AtBeginDocument{%

```

```

1646 \ltx@ifpackageloaded{hyperref}{
1647   \def\coppe@bibEnd{%
1648     \immediate\write\@auxout{%
1649       \string\newlabel{bib:end}{\arabic{page}}{\page.\arabic{page}}}%
1650   \def\coppe@bibBegin{%
1651     \immediate\write\@auxout{%
1652       \string\newlabel{bib:begin}{\arabic{page}}{\page.\arabic{page}}}%
1653   \def\coppe@mainBegin{%
1654     \immediate\write\@auxout{%
1655       \string\newlabel{front:pageno}{\Roman{page}}{\page.\roman{page}}}%
1656   \def\coppe@hasLof{%
1657     \immediate\write\@auxout{%
1658       \string\newlabel{cat:lofflag}{\il.;\page.\roman{page}}}%
1659 }{%
1660   \def\coppe@bibEnd{%
1661     \immediate\write\@auxout{%
1662       \string\newlabel{bib:end}{\arabic{page}}}%
1663   \def\coppe@bibBegin{%
1664     \immediate\write\@auxout{%
1665       \string\newlabel{bib:begin}{\arabic{page}}}%
1666   \def\coppe@mainBegin{%
1667     \immediate\write\@auxout{%
1668       \string\newlabel{front:pageno}{\Roman{page}}}%
1669   \def\coppe@hasLof{%
1670     \immediate\write\@auxout{%
1671       \string\newlabel{cat:lofflag}{\il.;\page.\roman{page}}}%
1672 }%
1673 }
1674 % The reference list is now produced by biblatex's \printbibliography (see the
1675 % bibliography setup after babel, above). The former thebibliography
1676 % redefinition and its \bibindent are no longer used and have been removed; the
1677 % bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
1678 % \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.

```

longquote (env.)

```

1679 \newlength{\recuolongquote}%
1680 \setlength{\recuolongquote}{4cm}%
1681 \newenvironment*{longquote}[1][default]{%
1682   \list{}%
1683   \footnotesize%
1684   \addtolength{\leftskip}{\recuolongquote}%
1685   \item[]%
1686   \singlespacing%
1687   \ifthenelse{\not\equal{#1}{default}}{\itshape\selectlanguage{#1}}{}%
1688 }{\endlist}%

1689 \newenvironment{theglossary}{%
1690   \if@two column%
1691     \@restonecoltrue\onecolumn%
1692   \else%
1693     \@restonecolfalse%
1694   \fi%
1695   \@mkboth{\MakeUppercase\glossaryname}%
1696   {\MakeUppercase\glossaryname}%

```

```

1697 \chapter*{\glossaryname}%
1698 \addcontentsline{toc}{chapter}{\glossaryname}
1699 \list{}
1700 {\setlength{\listparindent}{0in}%
1701 \setlength{\labelwidth}{1.0in}%
1702 \setlength{\leftmargin}{1.5in}%
1703 \setlength{\labelsep}{0.5in}%
1704 \setlength{\itemindent}{0in}}%
1705 \sloppy}%
1706 {\if@restonecol\twocolumn\fi%
1707 \endlist}
1708 %
1709 \renewenvironment{theindex}{%
1710 \if@twocolumn
1711 \@restonecolfalse
1712 \else
1713 \@restonecoltrue
1714 \fi
1715 \twocolumn[\@makeschapterhead{\indexname}]{%
1716 \mkboth{\MakeUppercase\indexname}%
1717 {\MakeUppercase\indexname}%
1718 \thispagestyle{plain}\parindent\z@
1719 \addcontentsline{toc}{chapter}{\indexname}
1720 \parskip\z@ \@plus .3\p@ \relax
1721 \columnseprule \z@
1722 \columnsep 35\p@
1723 \let\item\@idxitem}
1724 {\if@restonecol\onecolumn\else\clearpage\fi}
1725 \if@english
1726 \newcommand\listabbreviationname{List of Abbreviations and Acronyms}
1727 \newcommand\listsymbolname{List of Symbols}
1728 \newcommand\glossaryname{Glossary}
1729 \else
1730 \newcommand\listabbreviationname{Lista de Abreviaturas e Siglas}
1731 \newcommand\listsymbolname{Lista de Símbolos}
1732 \newcommand\glossaryname{Glossário}
1733 \fi
1734 %
1735 \newcommand\local@advisorstring{Orientador}
1736 \newcommand\foreign@advisorstring{Advisor}
1737 \ifthenelse{\boolean{maledoc}}{%
1738 \newcommand\local@approvedname{Examinado por}%
1739 }{%
1740 \newcommand\local@approvedname{Examinada por}%
1741 }
1742 \newcommand\local@universityname{Universidade Federal do Rio de Janeiro}
1743 \newcommand\local@deptstring{Programa}
1744 \newcommand\foreign@deptstring{Department}
1745 \newcommand\local@cityname{Rio de Janeiro}
1746 \newcommand\local@statename{RJ}
1747 \newcommand\local@countryname{Brasil}
1748 %
1749 \newcommand\frontcover@maintext{
1750 \sloppy\nohyphens{\local@doctype\ de \@degree\

```



```

1751 \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
1752 ao Programa de P{\ ' o}s-gradua{\c c}{\~ a}o em \local@deptname,
1753 COPPE, da \local@universityname, como parte dos requisitos
1754 necess{\ ' a}rios {\ ' a} obten{\c c}{\~ a}o do t{\ ' i}tulo de
1755 \local@degname\ em \local@deptname.}
1756 }
1757 %
1758 \newcommand\frontpage@maintext{
1759 \noindent {\MakeUppercase\local@doctype}
1760 \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
1761 \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
1762 DE P{\ ' O}S-GRADUA{\c C}{\~ A}O E PESQUISA DE ENGENHARIA DA
1763 UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
1764 NECESS{\ ' A}RIOS PARA A OBTEN{\c C}{\~ A}O DO GRAU DE
1765 {\MakeUppercase\local@degname} EM CI{\^ E}NCIAS EM
1766 {\MakeUppercase\local@deptname.\par}}%
1767 }
1768 %
1769 \newcommand\frontpage@bottomtext{%
1770 \begin{center}
1771 {\MakeUppercase{\local@cityname, \local@statename\ -- \local@countryname}}\par
1772 {\MakeUppercase\local@monthname\ DE \number\year}
1773 \end{center}%
1774 }
1775 %
1776 \newcommand\abstract@toptext{%
1777 \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}
1778 \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
1779 \sloppy\nohyphens{{\ ' a} COPPE/UFRJ como parte dos requisitos
1780 necess{\ ' a}rios para a obten{\c c}{\~ a}o do grau de
1781 \local@degname\ em Ci{\^ e}ncias (\@degree)}
1782 }
1783 \newcommand\foreignabstract@toptext{%
1784 \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to
1785 COPPE/UFRJ as a partial fulfillment of the requirements for the
1786 degree of \foreign@degname\ of Science (\@degree)}
1787 }
1788 %
1789 </class>
1790 <*glossary>
1791 actual '='
1792 quote '!'
1793 level '>'
1794 %%% delim_0 " , p. "
1795 delim_0 "\\dotfill "
1796 lethead_flag 0
1797 headings_flag 0
1798 preamble
1799 "\n\\begin{theglossary}\n \\makeatletter"
1800 postamble
1801 "\n \\end{theglossary}\n"
1802 </glossary>
1803 <*class>

```

```

1804 \newcommand{\annex}{
1805     \iflanguage{brazilian}
1806         {\renewcommand{\appendixname}{Anexo}}
1807         {\renewcommand{\appendixname}{Annex}}
1808     \appendix
1809     \@coppeannextrue
1810 }
1811 \end{class}

```

BibLaTeX style files

Generated from this .dtx by coppe.ins (one docstrip module each).

```

1812 \begin{dbx}
1813 \ProvidesFile{coppe.dbx}[2026/05/23 v3.8 CoppeTeX ABNT datamodel]
1814 % Custom entry types beyond biblatex's standard datamodel (games, TV, maps).
1815 % map = cartographic document (NBR 6023 4.2.12); biblatex has no native @map.
1816 % The remaining ABNT "exotic" types (legislation, jurisdiction, legal, patent,
1817 % standard, music, audio, image, artwork, video, dataset, letter, periodical,
1818 % performance) are already in biblatex's standard datamodel.
1819 \DeclareDatamodelEntrytypes{game,videogame,boardgame,tvshow,map}
1820 % Custom fields.
1821 \DeclareDatamodelFields[type=field,datatype=literal]{course}          % "(... em <course>)"
1822 \DeclareDatamodelFields[type=field,datatype=literal]{coppedegree}    % degree word
1823 \DeclareDatamodelFields[type=field,datatype=literal]{platform}       % videogame platform
1824 \DeclareDatamodelFields[type=field,datatype=literal]{scale}          % map scale (escala)
1825 \DeclareDatamodelFields[type=list,datatype=literal]{artist}          % game/illustration art
1826 \DeclareDatamodelFields[type=list,datatype=literal]{director}        % movie/tvshow/video
1827 \DeclareDatamodelFields[type=list,datatype=literal]{producer}
1828 \DeclareDatamodelFields[type=field,datatype=literal]{caderno}        % newspaper section
1829 \DeclareDatamodelEntryfields{course,coppedegree}
1830 \DeclareDatamodelEntryfields[game,videogame,boardgame,software,movie,tvshow,video]{%
1831     {artist,platform,director,producer,version}}
1832 \DeclareDatamodelEntryfields[map]{scale}
1833 \DeclareDatamodelEntryfields[article]{caderno}
1834 \end{dbx}
1835 \begin{bbx}
1836 \ProvidesFile{coppe.bbx}[2026/05/23 v3.8 CoppeTeX ABNT bibliography style]
1837 % The reference list is single-spaced (ABNT 2.4) via \singlespacing in
1838 % \AtBeginBibliography. coppe.cls already loads setspace, but the style must work
1839 % standalone too (e.g. \usepackage[bibstyle=coppe]{biblatex} in a plain article),
1840 % so load setspace here as well (idempotent).
1841 \RequirePackage{setspace}
1842 % ABNT reference list keeps the date at the end (imprenta), same for author-date
1843 % and numeric modes, so we derive from the standard style, not authoryear.
1844 \RequireBibliographyStyle{standard}
1845
1846 % Note: standard style never dashes repeated authors, which already matches the
1847 % post-2020 ABNT rule (repeat the entry in full), so no 'dashed' option is needed.
1848 \ExecuteBibliographyOptions{%
1849     maxbibnames=99,          % ABNT 4.3.1.3: list all authors by default
1850     giveninits=false,
1851     uniquename=false,
1852     uniquelist=false,

```

```

1853 labeldateparts=true, % needed for author-date extradate (1999a/1999b)
1854 sorting=nyt,
1855 }
1856
1857 % Language strings (Disponível em / Acesso em / In, etc.)
1858 \DeclareLanguageMapping{brazilian}{brazilian-coppe}
1859 \DeclareLanguageMapping{english}{english-coppe}
1860
1861 % --- Names: all reversed "FAMILY, Given", separated by ";" in the list ---
1862 \DeclareNameAlias{author}{family-given}
1863 \DeclareNameAlias{editor}{family-given}
1864 \DeclareNameAlias{translator}{family-given}
1865 \DeclareNameAlias{default}{family-given}
1866 \DeclareDelimFormat[bib,biblist]{multinamedelim}{\addsemicolon\space}
1867 \DeclareDelimFormat[bib,biblist]{finalnamedelim}{\addsemicolon\space}
1868
1869 % Uppercase the surname (family + prefix) only inside the bibliography list;
1870 % citations keep normal caps (post-2020 rule).
1871 \AtBeginBibliography{%
1872   \singlespacing % entries single-spaced internally (2.4)
1873   \setlength{\bibitemsep}{\baselineskip}% one blank line between entries (2.4)
1874   \renewcommand*{\mkbibnamefamily}[1]{\MakeUppercase{#1}}%
1875   \renewcommand*{\mkbibnameprefix}[1]{\MakeUppercase{#1}}%
1876 }
1877
1878 % --- Titles: bold for whole-document types; plain for parts; entry-by-title
1879 % (no author/editor) -> not bold, first word in CAPS (ABNT 4.2). ---
1880 % \coppeUCtitle uppercases the first word of the title. The delimited macros
1881 % split off the first word; \coppeUCtitleB strips the trailing-space sentinel
1882 % so no spurious space is left before the following punctuation.
1883 \protected\def\coppeUCtitle#1{\coppeUCtitleA#1 \coppeUCstop}
1884 \protected\def\coppeUCtitleA#1 #2\coppeUCstop{%
1885   \MakeUppercase{#1}\ifblank{#2}{ }\space\coppeUCtitleB#2\coppeUCstop}
1886 \protected\def\coppeUCtitleB#1 \coppeUCstop{#1}
1887 \DeclareFieldFormat{title}{%
1888   \ifboolexpr{ test {\ifnameundef{labelname}} }
1889     {#1}%
1890     {\mkbibbold{#1}}}
1891 % First word of the title in CAPS for entry-by-title (ABNT 4.2). The titlecase
1892 % format receives the RAW title text (the title format above only sees the
1893 % field-printing commands), so \coppeUCtitle can split off and uppercase the
1894 % first word here.
1895 \DeclareFieldFormat{titlecase}{%
1896   \ifboolexpr{ test {\ifnameundef{labelname}} }
1897     {\coppeUCtitle{#1}}%
1898     {#1}}
1899 \DeclareFieldFormat[article,incollection,inbook,inproceedings,suppperiodical,review]{title}{%
1900   \DeclareFieldFormat[thesis]{title}{\mkbibbold{#1}}
1901 % biblatex's standard style quotes patent (and unpublished) titles; ABNT bolds
1902 % the title of a whole document, so override those back to bold.
1903 \DeclareFieldFormat[patent]{title}{\mkbibbold{#1}}
1904 % Legal documents (NBR 6023 4.2.6): the law's identification/ementa is NOT
1905 % highlighted; the entry leads with the jurisdiction (entity author, CAPS).
1906 \DeclareFieldFormat[legislation,jurisdiction,legal]{title}{#1}

```

```

1907 \DeclareFieldFormat{journaltitle}{\mkbibbold{#1}}
1908 \DeclareFieldFormat{booktitle}{\mkbibbold{#1}}
1909 \DeclareFieldFormat{maintitle}{\mkbibbold{#1}}
1910
1911 % --- Pages: ABNT uses "p." (also for ranges); theses use "f." for pagetotal ---
1912 \DeclareFieldFormat{pagetotal}{#1\addnbspace p\adddot}
1913 \DeclareFieldFormat[thesis]{pagetotal}{#1\addnbspace f\adddot}
1914 \DeclareFieldFormat{pages}{p\adddot\addnbspace#1}
1915
1916 % --- Periodical volume / number prefixes ---
1917 \DeclareFieldFormat[article]{volume}{v\adddot\addnbspace#1}
1918 \DeclareFieldFormat[article]{number}{n\adddot\addnbspace#1}
1919
1920 % --- "In:" before host title (incollection/inproceedings/inbook) ---
1921 \renewbibmacro*{in:}{\printtext{\bibstring{in}\addcolon\space}}
1922
1923 % --- Online: "Disponível em: <url>." + "Acesso em: <date>." via lbx strings ---
1924 \DeclareFieldFormat{url}{\bibstring{availablefrom}\addcolon\space\url{#1}}
1925
1926 % --- ABNT article driver: no "In:"; Journal, Local, v., n., p., date, DOI/URL.
1927 %     Newspapers (entrysubtype=newspaper, set by the @artigodejornal source map)
1928 %     differ: the newspaper name (journaltitle) and city, then either
1929 %         "<date>. <caderno>, p. <pages>"      (when a caderno/section is given), or
1930 %         "p. <pages>, <date>"                  (page before date, when none),
1931 %     per NBR 6023 §4.2.3.6. ---
1932 \DeclareBibliographyDriver{article}{%
1933   \usebibmacro{bibindex}%
1934   \usebibmacro{begentry}%
1935   \usebibmacro{author/translator+others}%
1936   \setunit{\labelnamepunct}\newblock
1937   \usebibmacro{title}%
1938   \newunit\newblock
1939   \printfield{journaltitle}%
1940   \setunit{\addcomma\space}\printlist{location}%
1941   \iffieldequalstr{entrysubtype}{newspaper}
1942     {\iffieldequalstr{caderno}
1943       {\setunit{\addcomma\space}\printfield{pages}%
1944         \setunit{\addcomma\space}\printdate}%
1945       {\setunit{\addcomma\space}\printdate%
1946         \setunit{\adddot\space}\printfield{caderno}%
1947         \setunit{\addcomma\space}\printfield{pages}}}%
1948   {\setunit{\addcomma\space}\printfield{volume}%
1949     \setunit{\addcomma\space}\printfield{number}%
1950     \setunit{\addcomma\space}\printfield{pages}%
1951     \setunit{\addcomma\space}\printdate}%
1952   \newunit\newblock
1953   \printfield{doi}%
1954   \newunit\newblock
1955   \usebibmacro{url+urldate}%
1956   \usebibmacro{finentry}}
1957
1958 % --- incollection: ABNT puts the host organizer before the booktitle, as
1959 %     "AUTOR. Parte. In: ORGANIZADOR (Org.). Booktitle. Local: Editora, ano,
1960 %     p. X-Y." The organizer role string is "Org./Orgs." (see the .lbx). ---

```

```

1961 \newbibmacro*{coppe:bookeditor}{%
1962   \ifnameundef{editor}
1963     {}
1964     {\printnames{editor}%
1965       \setunit*{\addspace}\printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
1966       \clearname{editor}%
1967       \setunit{\adddot\space}}
1968 \newbibmacro*{coppe:partdriver}{%
1969   \usebibmacro{bibindex}\usebibmacro{begentry}%
1970   \usebibmacro{author/translator+others}%
1971   \setunit{\labelnamepunct}\newblock
1972   \usebibmacro{title}%
1973   \newunit\newblock
1974   \usebibmacro{in:}%
1975   \usebibmacro{coppe:bookeditor}%
1976   \printfield{booktitle}%
1977   \newunit\newblock
1978   \printfield{edition}%
1979   \newunit\newblock
1980   \printlist{location}%
1981   \setunit*{\addcolon\space}\printlist{publisher}%
1982   \setunit*{\addcomma\space}\printdate%
1983   \newunit\newblock
1984   \printfield{pages}%
1985   \setunit{\addcomma\space}\printfield{chapter}%
1986   \newunit\newblock
1987   \usebibmacro{url+urldate}%
1988   \usebibmacro{finentry}}
1989 \DeclareBibliographyDriver{incollection}{\usebibmacro{coppe:partdriver}}
1990 \DeclareBibliographyDriver{inbook}{\usebibmacro{coppe:partdriver}}
1991
1992 % --- type field: resolve a known bibliography string (babel-aware), else literal ---
1993 \DeclareFieldFormat{type}{\ifbibstring{#1}{\bibstring{#1}}{#1}}
1994
1995 % --- Thesis: "<type> (em <course>)" using the custom course field (ABNT). ---
1996 \DeclareFieldFormat{course}{\mkbibparens{\iflanguage{brazilian}{em\addspace}{in\addspace}#1}}
1997 \DeclareBibliographyDriver{thesis}{%
1998   \usebibmacro{bibindex}\usebibmacro{begentry}%
1999   \usebibmacro{author}%
2000   \setunit{\labelnamepunct}\newblock
2001   \usebibmacro{title}%
2002   \newunit\newblock
2003   \printfield{type}\setunit*{\addspace}\printfield{course}%
2004   \newunit\newblock
2005   \printlist{institution}%
2006   \setunit*{\addcomma\space}\printlist{location}%
2007   \setunit*{\addcomma\space}\printdate%
2008   \newunit\newblock
2009   \printfield{pagetotal}%
2010   \newunit\newblock
2011   \printfield{note}%
2012   \newunit\newblock
2013   \usebibmacro{url+urldate}%
2014   \usebibmacro{finentry}}

```

```

2015 % --- Games / audiovisual: render the custom fields (artist, platform,
2016 %      director, producer, version) the generic misc driver ignores. ---
2017 \newbibmacro*{coppe:media}{%
2018   \iflistundef{artist}{\setunit{\addperiod\space}%
2019     \printtext{\iflanguage{brazilian}{Ilustra\c{c}\~ao}{Art}\addcolon\space}\printlist{artist}%
2020   \iffieldundef{platform}{\setunit{\addperiod\space}%
2021     \printtext{\iflanguage{brazilian}{Plataforma}{Platform}\addcolon\space}\printfield{platform}%
2022   \iflistundef{director}{\setunit{\addperiod\space}%
2023     \printtext{\iflanguage{brazilian}{Dire\c{c}\~ao}{Directed by}\addcolon\space}\printlist{director}%
2024   \iflistundef{producer}{\setunit{\addperiod\space}%
2025     \printtext{\iflanguage{brazilian}{Produ\c{c}\~ao}{Produced by}\addcolon\space}\printlist{producer}%
2026 \newbibmacro*{coppe:mediadriv}{%
2027   \usebibmacro{bibindex}\usebibmacro{begentry}%
2028   \usebibmacro{author/translator+others}%
2029   \setunit{\labelnamepunct}\newblock
2030   \usebibmacro{title}%
2031   \newunit\newblock
2032   \usebibmacro{coppe:media}%
2033   \newunit\newblock
2034   \printlist{publisher}%
2035   \setunit*{\addcomma\space}\printlist{location}%
2036   \setunit*{\addcomma\space}\printdate%
2037   \newunit\newblock
2038   \printfield{version}%
2039   \newunit\newblock
2040   \printfield{note}%
2041   \newunit\newblock
2042   \usebibmacro{url+urldate}%
2043   \usebibmacro{finentry}}
2044 \DeclareBibliographyDriver{game}{\usebibmacro{coppe:mediadriv}}
2045 \DeclareBibliographyDriver{videogame}{\usebibmacro{coppe:mediadriv}}
2046 \DeclareBibliographyDriver{boardgame}{\usebibmacro{coppe:mediadriv}}
2047 \DeclareBibliographyDriver{movie}{\usebibmacro{coppe:mediadriv}}
2048 \DeclareBibliographyDriver{tvshow}{\usebibmacro{coppe:mediadriv}}
2049 % Cartographic document (map, NBR 6023 4.2.12): like a book (author/title/
2050 % imprenta) plus the mandatory scale ("Escala 1:N").
2051 \DeclareFieldFormat{scale}{\bibstring{coppescale}\adnbspace#1}
2052 \DeclareBibliographyDriver{map}{%
2053   \usebibmacro{bibindex}\usebibmacro{begentry}%
2054   \usebibmacro{author/translator+others}%
2055   \setunit{\labelnamepunct}\newblock
2056   \usebibmacro{title}%
2057   \newunit\newblock
2058   \printlist{location}%
2059   \setunit*{\addcolon\space}\printlist{publisher}%
2060   \setunit*{\addcomma\space}\printdate%
2061   \newunit\newblock
2062   \printfield{scale}%
2063   \newunit\newblock
2064   \usebibmacro{url+urldate}%
2065   \usebibmacro{finentry}}
2066 % Patent (NBR 6023 4.2.5): inventor. \textbf{title}. Depositante: holder. type
2067 % number. date. The holder (depositante) is printed in normal case, not the
2068 % ALL-CAPS family treatment the bibliography gives the leading author.

```

```

2069 \newbibmacro*{coppe:depositor}{%
2070   \ifnameundef{holder}
2071     {}
2072     {\printtext{\bibstring{depositor}\addcolon\space}%
2073       \begingroup
2074         \def\mkbibnamefamily##1{##1}%
2075         \def\mkbibnameprefix##1{##1}%
2076         \printnames{holder}%
2077       \endgroup}}
2078 \DeclareBibliographyDriver{patent}{%
2079   \usebibmacro{bibindex}%
2080   \usebibmacro{begentry}%
2081   \usebibmacro{author/translator+others}%
2082   \setunit{\labelnamepunct}\newblock
2083   \usebibmacro{title}%
2084   \newunit\newblock
2085   \usebibmacro{coppe:depositor}%
2086   \newunit\newblock
2087   \printfield{type}%
2088   \setunit*{\addspace}\printfield{number}%
2089   \newunit\newblock
2090   \printdate%
2091   \newunit\newblock
2092   \usebibmacro{url+urldate}%
2093   \usebibmacro{finentry}}
2094 % Exotic ABNT types that biblatex's standard style leaves without a driver:
2095 % give each a baseline (book-like for published docs: norma/partitura; misc for
2096 % the rest) so they print. Per-type ABNT refinements can follow later.
2097 \DeclareBibliographyDriver{standard}{\usedriver{}{book}}
2098 \DeclareBibliographyDriver{music}{\usedriver{}{book}}
2099 % Legal documents (NBR 6023 4.2.6-4.2.8): jurisdiction/entity (CAPS author),
2100 % the law id + ementa (plain, not bold), then location and date.
2101 \newbibmacro*{coppe:legaldriver}{%
2102   \usebibmacro{bibindex}\usebibmacro{begentry}%
2103   \usebibmacro{author/translator+others}%
2104   \setunit{\labelnamepunct}\newblock
2105   \usebibmacro{title}%
2106   \newunit\newblock
2107   \printlist{location}%
2108   \setunit{\addcomma\space}\printdate%
2109   \newunit\newblock
2110   \usebibmacro{url+urldate}%
2111   \usebibmacro{finentry}}
2112 \DeclareBibliographyDriver{legislation}{\usebibmacro{coppe:legaldriver}}
2113 \DeclareBibliographyDriver{jurisdiction}{\usebibmacro{coppe:legaldriver}}
2114 \DeclareBibliographyDriver{legal}{\usebibmacro{coppe:legaldriver}}
2115 \DeclareBibliographyDriver{audio}{\usedriver{}{misc}}
2116 \DeclareBibliographyDriver{video}{\usedriver{}{misc}}
2117 \DeclareBibliographyDriver{software}{\usedriver{}{misc}}
2118 \DeclareBibliographyDriver{image}{\usedriver{}{misc}}
2119 \DeclareBibliographyDriver{artwork}{\usedriver{}{misc}}
2120 \DeclareBibliographyDriver{performance}{\usedriver{}{misc}}
2121 \DeclareBibliographyDriver{dataset}{\usedriver{}{misc}}
2122

```

```

2123 % --- biber source map: Portuguese synonyms (entry types + fields) + thesis family ---
2124 \DeclareSourcemap{%
2125   \maps[datatype=bibtex]{%
2126     % entry-type PT synonyms -> canonical English
2127     \map{\step[tourcesource=livro,typetarget=book]}
2128     \map{\step[tourcesource=artigo,typetarget=article]}
2129     \map{\step[tourcesource=artigodejornal,typetarget=article,final]\step[fieldset=entrysubtyp
2130     \map{\step[tourcesource=partedelivro,typetarget=incollection]}
2131     \map{\step[tourcesource=emlivro,typetarget=inbook]}
2132     \map{\step[tourcesource=verbete,typetarget=inreference]}
2133     \map{\step[tourcesource=anais,typetarget=proceedings]}
2134     \map{\step[tourcesource=trabalhodeevento,typetarget=inproceedings]}
2135     \map{\step[tourcesource=periodico,typetarget=periodical]}
2136     \map{\step[tourcesource=relatorio,typetarget=report]}
2137     \map{\step[tourcesource=relatoriotecnico,typetarget=report]}
2138     \map{\step[tourcesource>manual,typetarget>manual]}
2139     \map{\step[tourcesource=correspondencia,typetarget=letter]}
2140     \map{\step[tourcesource=patente,typetarget=patent]}
2141     \map{\step[tourcesource=diverso,typetarget=misc]}
2142     \map{\step[tourcesource=jogo,typetarget=game]}
2143     \map{\step[tourcesource=jogoeletronico,typetarget=videogame]}
2144     \map{\step[tourcesource=jogodetabuleiro,typetarget=boardgame]}
2145     \map{\step[tourcesource=filme,typetarget=movie]}
2146     \map{\step[tourcesource=serietv,typetarget=tvshow]}
2147     \map{\step[tourcesource=programadetelevisao,typetarget=tvshow]}
2148     % Exotic ABNT types (NBR 6023 4.2.5-4.2.14); targets are biblatex-native
2149     % except map (declared in cope.dbx).
2150     \map{\step[tourcesource=legislacao,typetarget=legislation]}
2151     \map{\step[tourcesource=atonomativo,typetarget=legislation]}
2152     \map{\step[tourcesource=jurisprudencia,typetarget=jurisdiction]}
2153     \map{\step[tourcesource=documentojuridico,typetarget=legal]}
2154     \map{\step[tourcesource=documentocivil,typetarget=legal]}
2155     \map{\step[tourcesource=norma,typetarget=standard]}
2156     \map{\step[tourcesource=partitura,typetarget=music]}
2157     \map{\step[tourcesource=iconografico,typetarget=image]}
2158     \map{\step[tourcesource=obradearte,typetarget=artwork]}
2159     \map{\step[tourcesource=tridimensional,typetarget=artwork]}
2160     \map{\step[tourcesource=cartografico,typetarget=map]}
2161     \map{\step[tourcesource=mapa,typetarget=map]}
2162     \map{\step[tourcesource=audiovisual,typetarget=video]}
2163     \map{\step[tourcesource=conjuntodedados,typetarget=dataset]}
2164     \map{\step[tourcesource=fasciculo,typetarget=supperperiodical]}
2165     \map{\step[tourcesource=trabalhoapresentado,typetarget=unpublished]}
2166     % thesis family: retype to thesis and preset the (babel-aware) type word
2167     \map{\step[tourcesource=tese,typetarget=thesis]}
2168     \map{\step[tourcesource=dissertacao,typetarget=thesis]}
2169     \map{\step[tourcesource=masterdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
2170     \map{\step[tourcesource=mscdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
2171     \map{\step[tourcesource=dissertacaomestrado,typetarget=thesis,final]\step[fieldset=type,fi
2172     \map{\step[tourcesource=dscthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=d
2173     \map{\step[tourcesource=tesedoutorado,typetarget=thesis,final]\step[fieldset=type,fieldval
2174     \map{\step[tourcesource=phdthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=p
2175     \map{\step[tourcesource=tesephds,typetarget=thesis,final]\step[fieldset=type,fieldvalue=phd
2176     \map{\step[tourcesource=tcc,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tcc]}

```



```

2177 % field PT synonyms (unaccented) -> canonical English
2178 \map{\step[fieldsource=periodico,fieldtarget=journaltitle]}
2179 \map{\step[fieldsource=revista,fieldtarget=journaltitle]}
2180 \map{\step[fieldsource=jornal,fieldtarget=journaltitle]}
2181 \map{\step[fieldsource=titulo,fieldtarget=title]}
2182 \map{\step[fieldsource=subtitulo,fieldtarget=subtitle]}
2183 \map{\step[fieldsource=autor,fieldtarget=author]}
2184 \map{\step[fieldsource=editora,fieldtarget=publisher]}
2185 \map{\step[fieldsource=local,fieldtarget=location]}
2186 \map{\step[fieldsource=ano,fieldtarget=year]}
2187 \map{\step[fieldsource=data,fieldtarget=date]}
2188 \map{\step[fieldsource=edicao,fieldtarget=edition]}
2189 \map{\step[fieldsource=paginas,fieldtarget=pages]}
2190 \map{\step[fieldsource=numero,fieldtarget=number]}
2191 \map{\step[fieldsource=tradutor,fieldtarget=translator]}
2192 \map{\step[fieldsource=organizador,fieldtarget=editor]}
2193 \map{\step[fieldsource=instituicao,fieldtarget=institution]}
2194 \map{\step[fieldsource=tipo,fieldtarget=type]}
2195 \map{\step[fieldsource=nota,fieldtarget=note]}
2196 \map{\step[fieldsource=dataacesso,fieldtarget=urldate]}
2197 \map{\step[fieldsource=curso,fieldtarget=course]}
2198 \map{\step[fieldsource=grau,fieldtarget=coppedegree]}
2199 \map{\step[fieldsource=artista,fieldtarget=artist]}
2200 \map{\step[fieldsource=plataforma,fieldtarget=platform]}
2201 \map{\step[fieldsource=diretor,fieldtarget=director]}
2202 \map{\step[fieldsource=produtor,fieldtarget=producer]}
2203 \map{\step[fieldsource=versao,fieldtarget=version]}
2204 \map{\step[fieldsource=escala,fieldtarget=scale]}
2205 }%
2206 }
2207
2208 % --- ABNT list layout: flush left, single spacing, blank line between entries ---
2209 \setlength{\bibhang}{0pt}
2210 \setlength{\bibitemsep}{\baselineskip}
2211
2212 </bbx>
2213 <*cbx>
2214 \ProvidesFile{coppe.cbx}[2026/05/23 v3.8 CoppeTeX ABNT author-date citations]
2215 % Author-date is the CoppeTeX 4.0 default. authoryear-comp gives compressed
2216 % author-year cites with normal-caps names (post-2020 rule) out of the box.
2217 \RequireCitationStyle{authoryear-comp}
2218
2219 % --- NBR 10520:2023 author separator -----
2220 % Parenthetical citations separate authors with ";" -> (Silva; Souza, 2020).
2221 % Narrative citations (\textcite, natbib \citete) keep comma + "e/and" ->
2222 % Silva, Souza e Oliveira (2020). The reference list keeps its own ";" via
2223 % coppe.bbx ([bib,biblist]); these contextless formats cover the cite side.
2224 \DeclareDelimFormat{multinamedelim}{\addsemicolon\space}
2225 \DeclareDelimFormat{finalnamedelim}{\addsemicolon\space}
2226 \DeclareDelimFormat{textcite}{multinamedelim}{\addcomma\space}
2227 \DeclareDelimFormat{textcite}{finalnamedelim}{\addspace\bibstring{and}\space}
2228
2229 % --- apud: citation of a citation (ABNT NBR 6023 §4.1.2.2 / NBR 10520) -----
2230 % ABNT lists only the work actually CONSULTED. So only the consulted work (#4, a

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2231 % .bib key) is cited and enters the reference list; the ORIGINAL work (#2 author,
2232 % #3 year) was not consulted and is given as plain text -- it is never a cite key,
2233 % so it is never added to the bibliography.
2234 % \citetapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
2235 % -> Orig Author (year) apud Consulted (yr[, page]) [narrative]
2236 % \citepapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
2237 % -> (Orig Author, year apud CONSULTED, yr[, page]) [parenthetical]
2238 % #1 = optional postnote (page/loc) on the consulted work; "apud" is latin -> italic.
2239 \DeclareRobustCommand{\citetapud}[4][]{%
2240 #2\space(#3)\space\mkbibemph{apud}\space\textcite[#1]{#4}}
2241 \DeclareRobustCommand{\citepapud}[4][]{%
2242 \mkbibparens{#2\addcomma\space#3\space\mkbibemph{apud}\space
2243 \citeauthor{#4}\addcomma\space\citeyear{#4}%
2244 \ifblank{#1}{ }\{\addcomma\space\mkpageprefix[pagination]{#1}\}}
2245 \</cbx>
2246 \<*\bxbr>
2247 \ProvidesFile{brazilian-coppe.lbx}[2026/05/23 v3.8 CoppeTeX brazilian strings]
2248 \InheritBibliographyExtras{brazilian}
2249 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
2250 \DeclareBibliographyStrings{%
2251 inherit = {brazilian},
2252 availablefrom = {{Dispon'\i vel em}{Dispon'\i vel em}},
2253 urlseen = {{Acesso em}{Acesso em}},
2254 in = {{In}{In}},
2255 editor = {{Organizador}{Org\addot}},
2256 editors = {{Organizadores}{Orgs\addot}},
2257 depositor = {{Depositante}{Depositante}},
2258 coppescale = {{Escala}{Escala}},
2259 mscdiss = {{Disserta\c{c}\~{a}o de Mestrado}{Disserta\c{c}\~{a}o de Mestrado}},
2260 dscthesi = {{Tese de Doutorado}{Tese de Doutorado}},
2261 phdthesis = {{Tese de Doutorado}{Tese de Doutorado}},
2262 tcc = {{Trabalho de Conclus\~{a}o de Curso}{Trabalho de Conclus\~{a}o de Curso}}
2263 }
2264 \</\bxbr>
2265 \<*\bxen>
2266 \ProvidesFile{english-coppe.lbx}[2026/05/23 v3.8 CoppeTeX english strings]
2267 \InheritBibliographyExtras{english}
2268 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
2269 \DeclareBibliographyStrings{%
2270 inherit = {english},
2271 availablefrom = {{Available from}{Available from}},
2272 urlseen = {{Access on}{Access on}},
2273 in = {{In}{In}},
2274 depositor = {{Depositor}{Depositor}},
2275 coppescale = {{Scale}{Scale}},
2276 mscdiss = {{M\addot Sc\addot\ Dissertation}{M\addot Sc\addot\ Diss\addot}},
2277 dscthesi = {{D\addot Sc\addot\ Thesis}{D\addot Sc\addot\ Thesis}},
2278 phdthesis = {{Ph\addot D\addot\ Thesis}{Ph\addot D\addot\ Thesis}},
2279 tcc = {{Undergraduate Thesis}{Undergraduate Thesis}},
2280 }
2281 \</\bxen>
2282 \<*\numbbx>
2283 \ProvidesFile{coppe-numeric.bbx}[2026/05/24 v3.8 CoppeTeX ABNT numeric bibliography]
2284 % Numeric reference list for the ABNT numeric system (NBR 10520). It reuses the

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2285 % full coppe ABNT style (drivers, CAPS surnames, bold titles, "In:", p./f.,
2286 % the PT->EN source map and the language mappings) and only adds the [n] list
2287 % label and the numeric sorting on top.
2288 \RequireBibliographyStyle{coppe}
2289
2290 % ABNT numeric system lists references in the order they are first cited
2291 % (NBR 10520), i.e. unsorted. Override with sorting=nty for an alphabetical
2292 % numbered list. 'labelnumber' generates the [n] field.
2293 \ExecuteBibliographyOptions{labelnumber,sorting=none}
2294
2295 % Bracketed labels, both in the list and for any shorthand entries.
2296 \DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}
2297 \DeclareFieldFormat{shorthandwidth}{\mkbibbrackets{#1}}
2298
2299 % Reference list driven by the [n] label (copied from biblatex's numeric.bbx),
2300 % keeping the ABNT blank line between entries via \bibitemsep set in coppe.bbx.
2301 \defbibenvironment{bibliography}
2302 {
2303   \list
2304     {\printtext[labelnumberwidth]{%
2305       \printfield{labelprefix}%
2306       \printfield{labelnumber}}}
2307     {\setlength{\labelwidth}{\labelnumberwidth}%
2308       \setlength{\leftmargin}{\labelwidth}%
2309       \setlength{\labelsep}{\biblabelsep}%
2310       \addtolength{\leftmargin}{\labelsep}%
2311       \setlength{\itemsep}{\bibitemsep}%
2312       \setlength{\parsep}{\bibparsep}}%
2313     {\renewcommand*{\makelabel}[1]{\hss#1}}
2314   {\endlist}
2315   {\item}
2316 }
2317 \numcbx
2318 \ProvidesFile{coppe-numeric.cbx}[2026/05/24 v3.8 CoppeTeX ABNT numeric citations]
2319 % Numeric citation system (ABNT NBR 10520): in-text references are bracketed
2320 % numbers [1], compressed and sorted within a single cite group. This is the
2321 % opt-in alternative to the author-date default (coppe.cbx); it pairs with
2322 % bibstyle=coppe-numeric. With the biblatex natbib option, \citep -> [1] and
2323 % \citet -> Author [1].
2324 \RequireCitationStyle{numeric-comp}
2325 \numcbx

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Acknowledgments

Thanks to all COPPET_EX users who have reported their experience with this class. We also thank to professor Fernando Lizarralde and Heiko Oberdiek for their helpful comments. The authors would like to thank the National Council for Scientific and Technological Development (CNPq) of Brazil.

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	General: COPPE _{TEX} release 2.0.	1		General: Extend abbrev to work like	
v2.1				symb and accept a sorting key	1
	General: COPPE _{TEX} release 2.1:		v3.4		
	Matching the new rules.	1		General: Some examples for	
	<code>\advisor</code> : Advisors, co-advisors,			figures, tables, longtables, etc.	1
	co-co-advisors, etc., all of them		v3.4.1		
	are simply considered advisors.	26		General: Fix english option to	
v2.2				consider brazilian a second	
	General: Matching new guidelines,			language, describe it in	
	including new logo.	1		document	25
v2.2.2			v3.5		
	General: Fixed some text constants			General: New command annex	47
	in .bib and documented it here.	1	v3.5.1		
v3.0				General: New command annex now	
	General: Added support for			supports brazilian or english	47